

Manhaj Admin (Principal) — Phase 1 Workflow Handover

Phase: Phase 1 (build now). The reformat of docs/admin_v1_build_handover.md into the shared archetype structure, folding in the founder comment refinements (X-series). Companion to docs/handover_phase1_teacher.md (the sick-leave→substitute chain starts here, in §6, and lands there as T6), the Parent and Student archetype docs, and governed by docs/handover-assets/_SHARED_SPEC.md . Design spec: docs/superpowers/specs/2026-06-15-admin-rebuild-design.md . **Who it's for:** anyone picking up the admin (principal) build — the engineer wiring server actions + RLS, the hybrid CTO sizing the AI cost and the CP-SAT service, or the (non-technical) sales lead who has school access and must source MoE templates, payment-portal details, and staffing lists.

How to read this: §0 is the orientation (who the principal is, the screen-at-a-glance table, and the full phasing snapshot incl. deferrals and scraps). §1–§8 are one section per Phase-1 admin area, each with a plain-English description, the embedded screenshot, who-can-do-what, a Mermaid workflow, the backend/data linkage (exact tables), mock data tables (NEW marked), deferrals with rationale, and open questions. §Z is the build & data handover.

Diagrams are **Mermaid** — they render as pictures on GitHub and in our PDF build. The admin works on the **web app** (Next.js on Vercel) in Phase 1. Comms are **email-only via Resend** — never "WhatsApp" in customer-facing copy.

0. Archetype summary

The Manhaj **admin is the principal** (and the admins/advisors they invite). They are the system-of-record owner: they **log in and manage the team, read the morning AI briefing, enter and maintain the school's core records** (students, teachers, parents, sections) one trusted record at a time, **run the timetable** and find daily substitutes, **triage at-risk students and the admissions/leaver lifecycle, approve staff sick-leave, and send parent communications + regulator reports**. Where the teacher's job is to *enter* the day's signals, the principal's job is to *act on the aggregate* and *keep the spine clean*. Everything the admin writes feeds the other three archetypes: enrolments and sections gate every teacher roster, applicants convert into students, approved sick-leave fires the teacher substitute sheet, and report templates drive the parent digest/invoice/comms.

This Phase-1 admin build is the **de-scope + write-enable pass**: turn the read-only demo into a real operations console. Bulk CSV import is deferred so the data stays clean; ad-hoc single-record entry comes first.

Phase-1 screens at a glance

Screen	Purpose	Key data it READS	Key data it WRITES	Links to
A1 · Auth & Manage Admins	Magic-link onboarding; principal invites/deactivates admins & advisors	school_admins NEW , invitations NEW	invitations NEW , school_admins NEW , audit_log	Every screen (role guard + RLS); Resend
A2 · Dashboard	Cached AI morning briefing + live count cards into each tab	ai_briefings NEW , live counts across spine	ai_briefings NEW , ai_usage_ledger	All tabs (drill-in)
A3 · Input Data (CRUD)	One trusted place to create/edit students, teachers, parents, section mapping	spine roster tables	students , teachers , parents , student_parents , student_enrollments , teacher_contracts , sections	A4 Faculty, A5 Students, A7 Schedule, A8 Reports
A4 · Faculty / Teacher HR	Roster, contracts, staffing category, hiring funnel, sick-leave approval queue	teachers , teacher_contracts , v_teacher_load	staff_absences NEW , staffing-category field NEW	A7 Schedule (coverage), Teacher T6 (substitute sheet)
A5 · Students (at-risk / applicants / leavers)	At-risk routing, re-enrolment funnel, leavers, applicant pipeline + convert	students , risk_flags NEW , student_enrollments , behaviour_notes	risk_flags NEW , applicants NEW , students.advisor_id NEW , students.withdrawn_reason NEW	Advisor board, Teacher app (homeroom owner), A8 Reports
A6 · Attendance	Attendance patterns/chronic-absentee view + teacher-sickness intake wiring	attendance_marks , staff_absences	(reads; sick-leave intake shared with A4)	Parent digest (P1), A4 queue, A5 at-risk
A7 · Schedule + auto-scheduling	Weekly timetable, daily substitute finder, CP-SAT auto-generate, NL constraints	teacher_section_subject , bell_periods , rooms , sections , staff_absences	timetable_slots , ai_usage_ledger , audit_log	Teacher T6 (sub handoff), every teacher roster
A8 · Reports	Parent-comms pipeline (draft→approve→send) + regulator reports (gated on MoE research)	comm_templates , comm_drafts , consent_records , report_archive NEW , v_assessment_oversight NEW (read-only view)	comm_drafts , report_archive NEW , regulatory_report_catalog NEW , audit_log	Parent P5 (invoice) + Parent P2/P3 (deferred) (comms), regulators, Sales review (X23)

Phasing snapshot

Item	Phase	Why (one line)
A1 Auth + Manage Admins	Phase 1	No real users without it; gates every other screen.
A2 Dashboard (cron AI briefing + count cards)	Phase 1	Cached, cheap, instant; the principal's daily entry point.
A3 Input Data — Students / Teachers / Parents / Section Mapping CRUD	Phase 1	The core new write surface; dashboards are only as good as this data.
A4 Faculty HR — roster, contracts, staffing category, sick-leave approval	Phase 1	Sick-leave approval is the trigger for the teacher substitute sheet.
A5 Students — at-risk routing, re-enrolment, leavers, applicant pipeline	Phase 1	The biggest SIS gap (admissions/enrolment write-path).
A6 Attendance — patterns view + sick-leave intake wiring	Phase 1	Reads <code>attendance_marks</code> (teacher-written); intake shared with A4.
A7 Schedule + CP-SAT auto-scheduling + NL constraints + substitute finder	Phase 1	Necessity, not deferred; the heaviest engineering item.
A8 Reports — parent-comms pipeline + report archive/tracking table	Phase 1 (comms)	Draft→approve→send is buildable now; archive + retention (X16) is P1.
A8 Regulatory reports (MoE)	Phase 1 intent · BLOCKED	Build gated on sourcing MoE templates/content/deadlines (research first).
Assessment/exam oversight view (X13)	Phase 1 (read-only)	Admin oversight table over <code>assessments / assessment_results</code> ; no new write path.
Report-view validation with school (X23)	Phase 1 task	Sales/CTO review report views with the school for custom/better views.
Bulk CSV / roster import	Phase 2	Deliberately capped in P1 to keep the spine clean.
Performance Composite (teacher scoring)	Phase 2–3 / demo-PDF only	Scrapped from live; kept as a PDF mockup.
Fee delinquency / AR console (admin-side)	Phase 2 (admin) · P5 (parent)	Invoices/payment live in Parent P5 (X20 moved); admin AR view is later.
Safeguarding / incident log (full case mgmt)	Phase 2	P1 records concern notes via <code>behaviour_notes / risk_flags</code> ; full case workflow later.
Reception / gate pickup feed	Phase 2	Operational nicety; needs a gate/check-in data source not in P1.
Parent-satisfaction analytics	Phase 2	Depends on a DEFERRED parent-feedback surface (X17); Parent P1 (weekly digest) captures no evals, so there is no Phase-1 data source.
Teacher↔parent comms drafts	Moved to Parent P2/P3 (deferred) (X21)	Comms belong to the Parent archetype, not admin.
Finance module (ledger/tables)	Later phase	No finance tables in the spine; separate initiative.

Founder confirmations folded in: X19 — all data requirements + the backend build for this admin scope are **confirmed/agreed**. **X20** → **Parent P5** (invoices) and **X21** → **Parent P2/P3 (deferred)** (comms): both **moved to the Parent archetype** — not built here. **X16/X17/X18** land in A8 (report archive + retention, satisfaction dependency, eval storage + the role a sign-up gives a teacher) — see those sections.

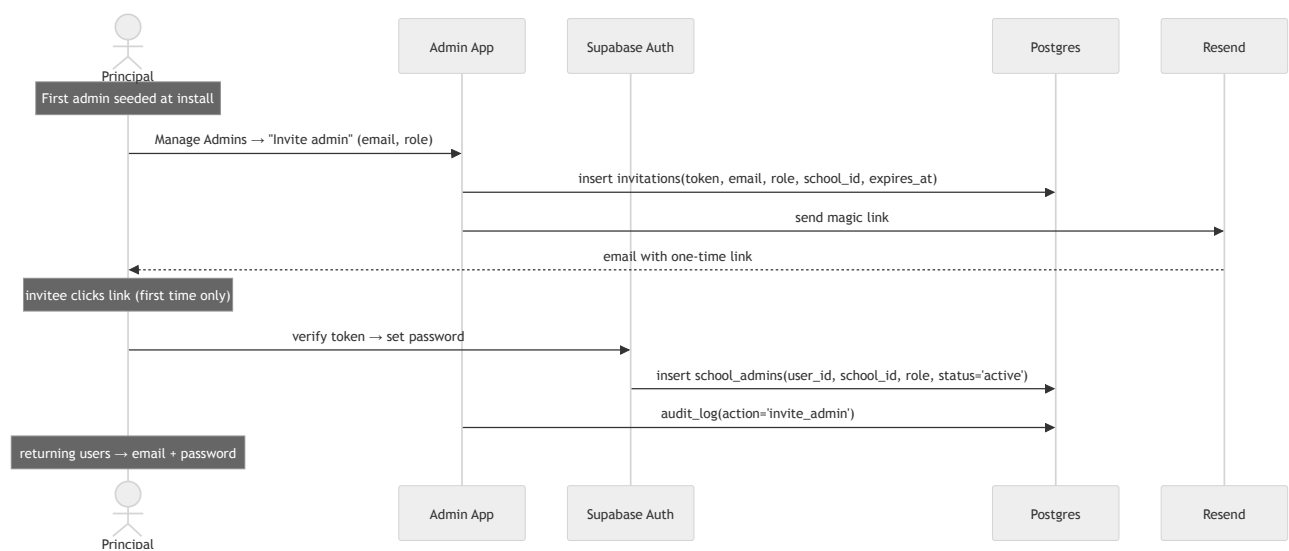
1. A1 — Authentication & Manage Admins

What it is. Users don't self-sign-up. The first admin (the principal) is seeded at install; that admin invites everyone else — admins and advisors — by email. Each invitee gets a one-time **magic link**, sets a password, and thereafter logs in with **email + password** (X2). Manage Admins is the settings-level page where the principal lists, invites, sets the role of, and deactivates team members.

Visual. (No dedicated mockup — this is the auth shell; Manage Admins surfaces as a settings table.)

Who uses it & permissions. Only `role='admin'` (the principal and senior admins) can invite or deactivate. Advisors can log in but cannot manage the team or touch faculty contracts. The **role guard** (shared `packages/auth`) reads `school_admins.role` + the `school_id` JWT claim on every request; Postgres **RLS** enforces the same `school_id` as defence in depth.

Workflow.



Backend & data linkage.

- `inviteAdmin(email, role)` server action writes one `invitations` (NEW) row and calls **Resend**. Guarded so only `role='admin'` can invite.
- The magic-link callback verifies the token against `invitations`, creates the `school_admins` (NEW) membership row (links to Supabase `auth.users`), and stamps `audit_log` (existing).
- Deactivation flips `school_admins.status` and writes `audit_log` — the row is retained, not deleted, for the compliance trail.
- The on-screen Manage Admins table maps one row to one `school_admins` record (joined to `auth.users` for last-active).

Data tables.

NEW `invitations` — pending invites:

email	role	school_id	status	expires_at
n.alkindi@school.om	advisor	(Al-Noor)	pending	2026-06-24T17:00:00Z
s.swart@school.om	admin	(Al-Noor)	accepted	2026-06-20T09:00:00Z

NEW `school_admins` — team membership + role:

user_id	role	status	invited_by	created_at
(Principal · A. Said)	admin	active	(seed)	2026-06-01T08:00:00Z
(S. Swart)	admin	active	A. Said	2026-06-20T09:12:00Z
(N. Al-Kindi)	advisor	pending	A. Said	2026-06-21T16:40:00Z

Deferred to P2/P3. None — A1 is Phase 1.

Open questions.

- Advisor as an **admin sub-role** vs a **teacher with an advisor flag** — affects which app the advisor logs into (admin vs teacher). Confirm with the at-risk owner decision in §5.
- Magic-link expiry window (24h default) — confirm with the school's IT policy.

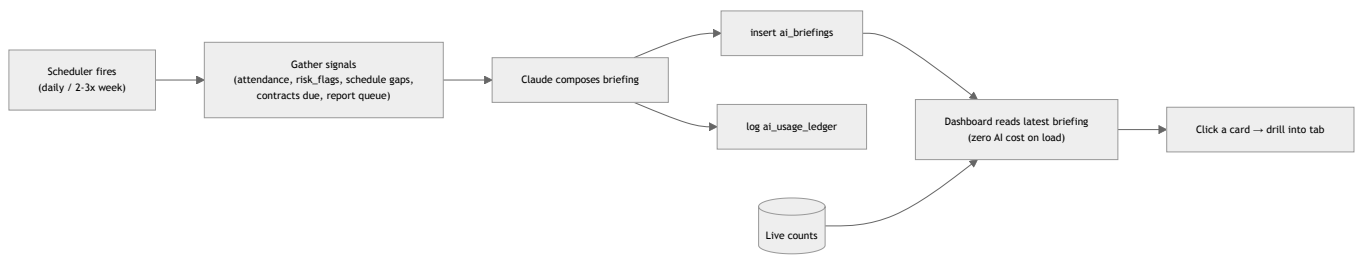
2. A2 — Dashboard

What it is. The principal's morning briefing: a short **AI-written summary** of what needs attention, plus live count cards (Faculty / Students / Attendance / Schedule / Reports) that link into each tab. The AI summary is **pre-computed on a schedule** (cron, daily or 2–3x/week — X3) and cached, so the page is instant and costs nothing on load.

Visual. (No dedicated mockup screenshot; the dashboard is the briefing header + count cards composed from the live tabs below.)

Who uses it & permissions. Admin/Principal (full). Advisors see a scoped version (their counts only via RLS).

Workflow.



Backend & data linkage.

- A scheduled job (**cron / Trigger.dev**) queries the spine for the day's signals, sends them to **Claude** (Haiku-class, background), and writes one row to **ai_briefings** (NEW: `school_id`, `scope`, `body`, `generated_at`). Cost logged to **ai_usage_ledger** (existing).
- The dashboard page **reads the latest ai_briefings row** and renders count cards from live queries against `attendance_marks` , `risk_flags` , `timetable_slots` , `teacher_contracts` , `comm_drafts` .
- Each card is a link into the corresponding tab — no logic, just navigation. A stale-briefing timestamp ("Updated {date}") shows if older than the cadence.

Data tables.

NEW ai_briefings — cached morning summary:

scope	body (excerpt)	generated_at
daily	"3 students crossed the at-risk threshold overnight (attendance). Mr Tariq is on approved sick-leave — G5B/G3A/G6C need cover. 2 re-enrolments still undecided in G9."	2026-06-21T05:00:00Z

ai_usage_ledger (existing) — one row per briefing run:

request_kind	model	input_tokens	output_tokens	cost_usd	is_background
dashboard_briefing	claude-haiku-4-5	2,400	320	0.0019	true

Deferred to P2/P3. None — A2 is Phase 1. (Per-advisor personalised briefings are a natural P2 extension once the at-risk owner routing in §5 is confirmed.)

Open questions.

- Dashboard cadence (X3 / D3): daily vs 2–3x/week — drives the cron schedule and the per-school AI cost.

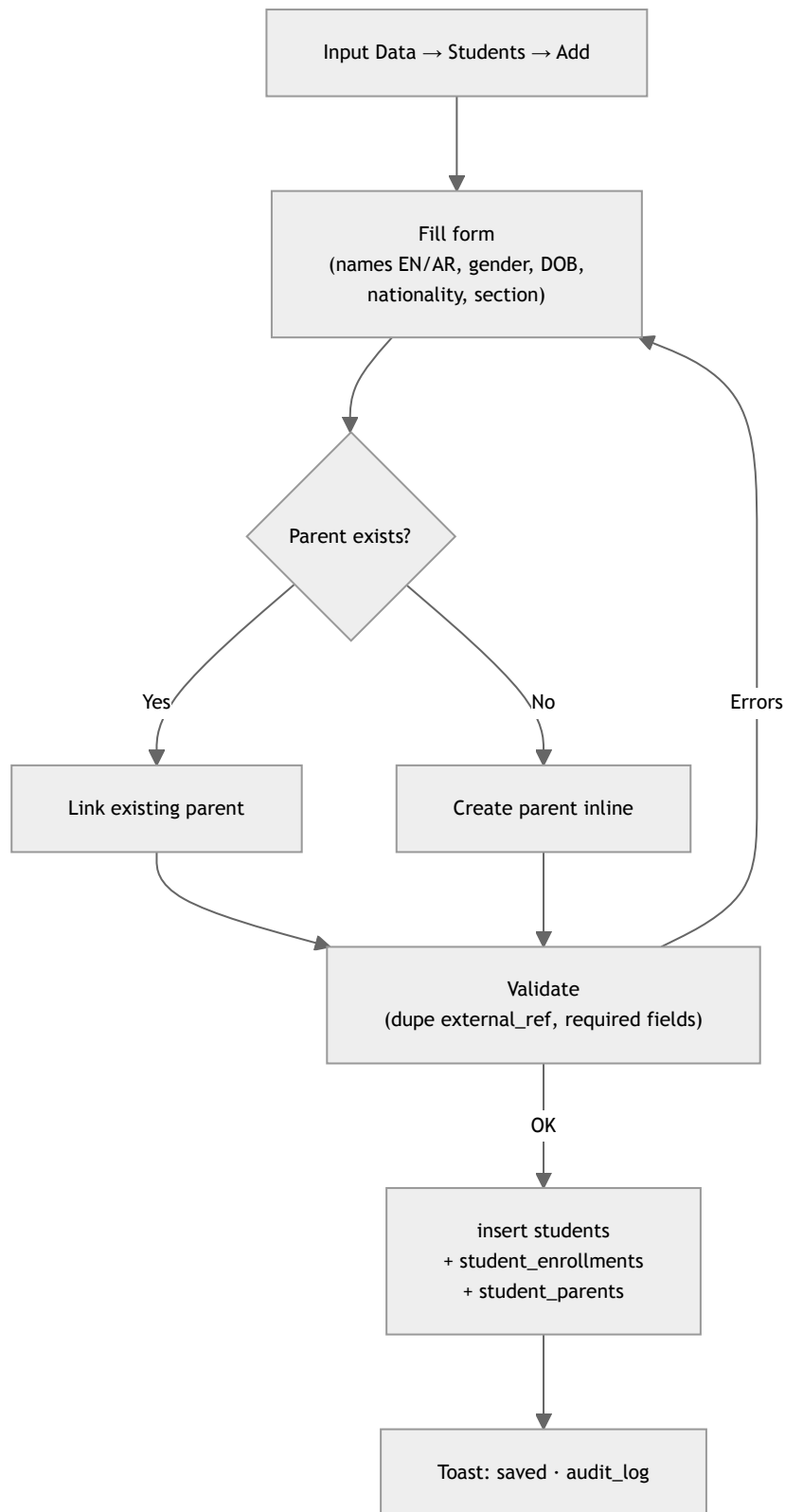
3. A3 — Input Data (ad-hoc CRUD) (only tab with sub-tabs)

What it is. One trusted place to create and edit the school's core records, **one record at a time**, with validation. Four sub-tabs: **Students, Teachers, Parents, Section Mapping** (X4: Input Data is the only tab allowed sub-tabs). Bulk CSV import is intentionally **switched off in P1** (X9) so the spine stays clean.

Visual. (No dedicated mockup; admissions/leaver records that originate here are visualised in §5 via `admin-admissions-pipeline` .)

Who uses it & permissions. Admin (full create/edit). Advisor is read-only on most sub-tabs. Section Mapping is admin-only and **gates Reports** (A8) — reports can't send for unmappped sections.

Workflow (add a student, with optional parent link). Create-teacher and create-parent are the **same shape**; one diagram, differences annotated below.



Create teacher = same shape, change: form fields are name/department/staffing category(NEW)/employment-status/contract; writes teachers + teacher_contracts (weekly_period_cap) + the load matrix teacher_section_subject ; no parent link. Feeds A7 Schedule (solver needs the load matrix). **Create parent = same shape, change:** fields are name/phone/email/preferred-lang/linked-student(s); writes parents + student_parents ; the validation that matters is duplicate email. Feeds A8 Reports (who receives comms) + consent. **Section Mapping = different:** confirms grade_level/label/stream/capacity and sets is_mapped=true , mapped_at , mapped_by on sections (columns from 006_section_mapping.sql). This **gates A8 Reports**.

- Each sub-tab is a list + an add/edit form backed by a typed server action: `createStudent / updateStudent` , `createTeacher / updateTeacher` , `createParent / updateParent` , `saveSectionMapping` .
- **Students** writes `students` , optionally `student_enrollments` (home section) and `student_parents` . A new student becomes immediately available to A5 (at-risk + re-enrolment).
- **Teachers** writes `teachers` + `teacher_contracts` + the **staffing-category** field (NEW). Feeds A7 (the solver reads `teacher_section_subject`).
- **Parents** writes `parents` + `student_parents` . Feeds A8 + consent.
- **Section Mapping** writes the human-confirmed columns on `sections` ; gates A8 sends.
- The bulk-import card renders but is **disabled** ("Coming soon" — X9 / D5).

Data tables.

`students` (existing + NEW columns introduced in §5) — created via Input Data:

full_name_en	external_ref	gender	date_of_birth	nationality	enrolled_on
Layla Al-Habsi (Grade 10)	STU-2451	F	2010-03-12	OM	2026-06-01
Khalil Al-Mahri (Grade 5)	STU-2452	M	2014-11-03	OM	2026-06-01

`teachers` (existing + NEW `staffing_category_id`):

full_name	primary_dept	staffing_category_id NEW	employment_status
Ms S. Swart	Maths	teaching_full	active
Mr Tariq Al-Bahri	Maths	teaching_full	active

`sections` (existing — confirmed via Section Mapping):

code	grade_level	stream	capacity	is_mapped	mapped_by
9A	9	regular	28	true	A. Said
11 AS	11	AS	22	true	A. Said

Deferred to P2/P3. Bulk CSV / roster import — capped in P1 to keep the spine clean. *Clean-linkage if built later*: a P2 importer should write through the **same** `createStudent / createTeacher / createParent` validation path (not a raw insert), so dedupe-by- `external_ref` and the `audit_log` trail are identical to ad-hoc entry — no separate code path to drift.

Open questions.

- Staffing-category list (X-series D7): does the school have the categories, or is defining them part of the work? (Sales to source.)
- Confirm `students.external_ref` is the dedupe key the school already uses.

4. A4 — Faculty / Teacher HR

What it is. The teacher system-of-record: who works here, what they teach, contract and workload, **staffing category**, a light **hiring funnel**, and the **sick-leave approval queue**. The old "Performance Composite" is removed from live (X5 / D6 → Phase 2). The sick-leave queue is the **trigger** for the teacher substitute sheet (Teacher doc §4 / T6).

Visual. Faculty roster + sick-leave approval is the admin side; the assigned-substitute flow is visualised in §7 below via the scheduler.

[Visual summary](#) [Backend & data](#) [The markup](#) [Open markup full screen](#)

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Today's schedule view with coverage status, AI-ranked substitute matching for absences, and a read-only master timetable. The killer feature is daily substitute finding — not the once-a-year timetable build.

www.elsevier.com/locate/jmb

Operations head, principal, or anyone managing daily school affairs.

RELIABILITY OF QUESTIONNAIRES

When a teacher calls in sick, automatically finds qualified substitutes, ranks by fit (subject competence + availability + load + dept + recency), assigns, and triggers the handoff sheet (TS).

NEW FEATURES

- View tabs (Today / This week / Weather timetable / Cover history)
- Day state: classes, teachers absent, cover assigned, open gaps
- Absent teacher block showing each affected period
- Per-period candidate ranking with match score (subject + free + load + dept)
- One-click assign = triggers handoff sheet to substitute (TE)
- Same substitute all-day preference reflected in scoring
- Master timetable grid (read-only) with SUB-tagged covered cells
- Plan next week forward navigation feed to constraint solver

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Data inputs, backend services, AI rule, and privacy considerations.

INPUT DATA REQUIRED

- Master timetable (per teacher / per class / per period)
- Teacher subject competencies (who's qualified to teach what)
- Teacher availability today (free periods, plus absence flags)
- This-week and lifetime substitute-load per teacher (fairness)
- Department assignments
- Recent sub-assignments (to avoid over-using one person)

图 20-1-1 所示为 2000 年 12 月 1 日 00:00 时, 某站测得的 10 min 平均风速为 10 m/s, 求该站 10 min 平均风速的 10 min 平均风速的 10 min 平均风速。

- Teacher subject competency table
- Daily availability state (three periods derived from timetable, modified by absences)
- Subject matching SQL, with ranked scoring (subject hard-filter, then weighted soft factors)
- One-click assignment workflow that pushes to 76 handoff generator
- Same-timetable all-day preference logic in the linker
- Master timetable: stored relationally per (period + class + room + teacher)
- Plan next week: CP-SAT solver (3% in service) → separate build for the same use case

1000000

Light for the daily matcher (deterministic ranking). Heavy for the annual timetable solver (CP-SAT constraint optimisation – separate feature).

PRIVATE & ACCESS

Admin role. Teacher absence reasons may be restricted (ask / personal don't need to be shared with all staff)

6. Feed schedule

Embedded below — or open full-screen [P](#) to interact with it directly.

Mathia

International School of Oman

Dashboard
Students
Faculty
Schedule
Reporting

Operations Mr. Sam

Today's schedule

Sunday 31 May 2020 - check classes, find substitutes, view the full school day.

Today

This week

Master timetable

Cover history

M

1 teacher absent today, 4 periods needing cover - all 4 now assigned. Ms. Fatima covers Mr. Tariq's whole day - she's accepted and the handoff sheet is in her inbox. No other gaps. School is fully staffed.
(seeing who requested substitute next night) Mr. Sam confirmed at 22:16

Classes Today	Teachers Absent	Covers Assigned	Open Gaps
47 across 35 sections	1 Mr. Tariq, sick leave	4 / 4 All periods covered	0 School fully staffed

TODAY'S SUBSTITUTIONS

TB

Mr. Tariq Al-Sayfi - Maths
Out today - sick leave reported 21:30 last night - returns tomorrow

2 teaching periods
4 cover candidates

08:00

P1 - 45 min

G5B Maths Room 10 - 20 students
[SUBJECT] [SECTION] [CAPTION]
Topic: Equivalent Fractions Intro - lesson plan + materials in mail

Ms. Fatima Al-Rimal

search MR - accepted 22:16

✓

WHY MS. FATIMA - RANKED CANDIDATES

Change pick

Ms. Fatima Al-Rimal

MS ranked first - has best topic knowledge

3%

Mr. Khalid

Head of Maths - Has P1 only, High rank

7%

Mr. Omer

Science, not class at 08:00, already teaching 2nd

09:00

P2 - 45 min

Free period - no class

No cover needed

09:45

P3 - 45 min

G3A Maths Room 8 - 18 students
[SUBJECT] [SECTION] [CAPTION]
Topic: Multiplication tables review - practice lesson

Ms. Fatima Al-Rimal

Match TB - same sub all day - accepted 22:16

✓

10:45

P4 - 45 min

G5B Maths Room 10 - same class as P1
Topic: Equivalence practice (continues P3) - self-led

Ms. Fatima Al-Rimal

Match TB - continuity with P3 - accepted 22:16

✓

12:00

Lunch

Lunch - staff meals

12:45

P5 - 45 min

D5C Maths Room 14 - 24 students
[SUBJECT] [SECTION] [CAPTION]
Topic: Decimal fractions intro - video lesson on smartboard

Ms. Fatima Al-Rimal

Match TB - already sub cover - handoff sent

✓

13:30

P6 - 45 min

Free period

No cover needed

Handoff sheet - sent to Ms. Fatima at 22:16 last night - she'll email it at 00:42 today

View Handoff sheet

Message Ms. Fatima

Mark Mr. Tariq returned

Full-screen A

Print A

Print B

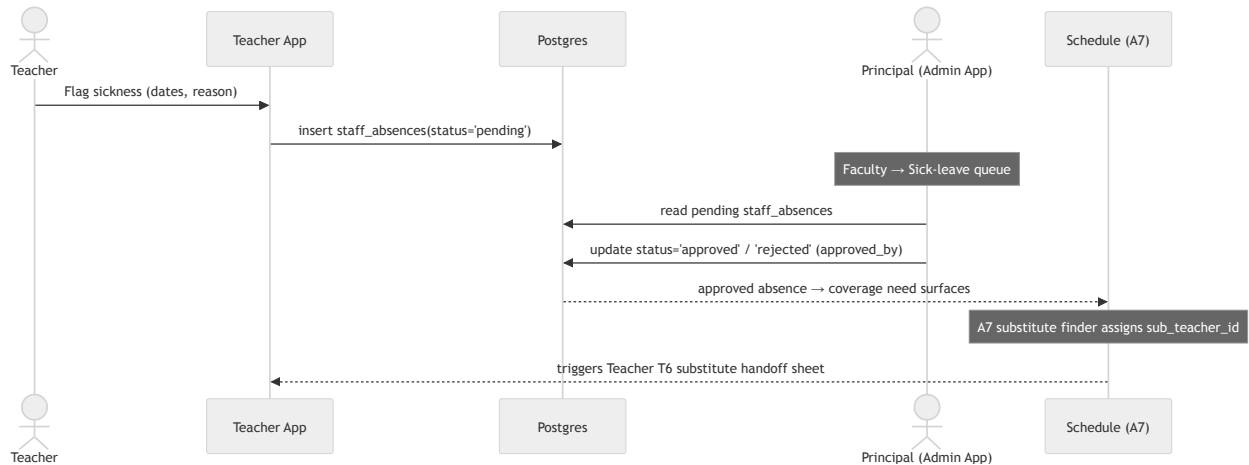
Print C

Print D



Who uses it & permissions. Admin (full); advisor read-only. Approve/reject on sick-leave is admin-only and guarded.

Workflow (staff sick-leave approval — the trigger for Teacher T6).



Backend & data linkage.

- Roster / detail** read `teachers` + `teacher_contracts` + the `v_teacher_load` view (assigned vs cap vs slack — already in `001_init.sql`).
- Staffing category (NEW)** is set on the teacher form (A3); categories drive workload grouping and later reporting. The category list is part of the work to define.
- Sick-leave:** teacher submits in the teacher app → `staff_absences` (NEW) row `status='pending'` → appears in the admin Faculty queue → the principal's approve/reject is an `update` server action (guarded) writing `approved_by` → an **approved** absence is read by A7 to surface the coverage need and, once a sub is assigned, fires the Teacher T6 handoff build.
- Hiring funnel** is a light pipeline view; in P1 it reads `teachers.employment_status` stages (`applicant` / `offer` / `active`) — a dedicated funnel table is a decision for the implementation plan.

Data tables.

NEW `staff_absences` — the sick-leave queue (trigger row for Teacher T6). `teacher` , `approved_by` , and `sub_teacher_id` show names as display stand-ins for the uuid FK.

teacher	start_date	end_date	reason	status	approved_by	sub_teacher_id
Mr Tariq Al-Bahri	2026-05-31	2026-05-31	sick	approved	A. Said	Ms Fatima Al-Hinai
Ms Noor Al-Saadi	2026-06-02	2026-06-03	sick	pending	—	—

`v_teacher_load` (existing view) — assigned vs cap:

teacher	weekly_period_cap	assigned	slack
Ms S. Swart	30	27	3
Mr Tariq Al-Bahri	30	30	0

Deferred to P2/P3. Performance Composite (teacher scoring) → Phase 2–3, kept as a demo PDF only. *Clean-linkage if built later:* it would read `rubric_scores` / `assessment_results` / `attendance_marks` already written by teachers — purely additive, no migration of A4 data.

Open questions.

- Staffing-category list source (shared with A3 open question).
- Hiring funnel: reuse `teachers.employment_status` stages vs a dedicated funnel table.
- Substitute assignment process — how/when a sub is named after approval (sales to confirm; feeds the A7 substitute finder + the T6 trigger point).

5. A5 — Students (at-risk routing · re-enrolment · leavers · applicants)

What it is. The student lifecycle and triage hub. Three jobs: (1) **route at-risk flags** to whoever owns the student so they get actioned; (2) manage **re-enrolment** and record **confirmed leavers**; (3) run a **new-applicant pipeline** and convert accepted applicants into enrolled students. The at-risk grading mechanism is kept as-is (D8) — the Phase-1 work is the **routing** (where a flag goes and where it surfaces).

Visual.

NA Nasser Al-Amri DOB Feb 2018 - contacted via Facebook	G3 - 2026/27	INQUIRY	3	Paid - Facebook ad	— unassigned	• Respond s24h	Claim →
FK Faisal Al-Kindi DOB Apr 2008 - current: TAISM	G12 - 2026/27	LOST	—	Open day	Dr. Patel	• Reason: timetable mismatch	View →

Click any row to open a detail panel — full timeline, documents, parent comms, and assessment scores.

14. Admissions + Re-enrolment

Admissions + Re-enrolment

Combined view of current student re-enrolment status for next year and new applicant pipeline. Both halves treated as one math problem: net enrolment = retained + new - leaving.

Combined view of current student re-enrolment status for next year and new applicant pipeline. Both halves treated as one math problem: net enrolment = retained + new - leaving.

Visual summary Backend & data The mockup

[Open mockup full-screen](#)

1. VISUAL SUMMARY

What this screen is

Combined view of current student re-enrolment status for next year and new applicant pipeline. Both halves treated as one math problem: net enrolment = retained + new - leaving.

WHO IT'S FOR

Principal + admissions officer.

WORKFLOW IT SUPPORTS

Solves "will we have a viable class next year?" The retention half catches non-renewals, the new applicant half fills the gap.

KEY FEATURES

- Headline state: current enrolled, re-enrolled count + %, pending decisions, confirmed leaving
- Re-enrolment families list (high-risk first, with reason & avg. recommended action)
- "Confirmed leaving - and why" breakdown (graduating / moving / switching schools)
- Year-over-year comparison ("1.8 pts vs same time last year")
- New applicant funnel by stage (Inquiry → Tour → Assessment → Interview → Offer → Enrolled)
- Per-stage conversion rate ("57 is looking at interview")
- Action queue: re-enrolment chasers + new applicant follow-ups + replacement-capacity forecast
- Detailed applicant table with state-stage detection and AI signal per applicant

2. BACKEND & DATA

What it takes to build this for real

Data inputs, backend services, AI role, and privacy considerations.

INPUT DATA REQUIRED

- Current student roster + re-enrolment confirmation status
- Leavers (with stated reason)
- Applicants table with stage timestamps
- Source attribution per applicant (advertising referral / open day / Facebook ad / etc.)
- Communication history per applicant (responsiveness)
- Last 3 years of historical funnel for base-rate comparison

BACKEND RULES REQUIRED

- Two parallel pipelines: re-enrolment workflow + new applicant CRM
- State-stage detection job (alert when applicant doesn't move for >30 days)
- Conversion-rate analytics per stage and per grade
- Cross-link to at-risk dashboard (high-risk re-enrolments flagged automatically)
- AI draft generation for follow-up messages (consent reminders, offer nudges)
- Year-over-year baseline storage for the comparison line

AI ROLE

Medium-heavy: AI drafts follow-up communications, identifies stuck applicants, predicts which offers are most likely to convert, ties retention risks to admissions viability.

PRIVACY & ACCESS

Admissions team + principal access. Applicant data subject to GDPR-equivalent regulations once collected.

3. THE MOCKUP

Live preview

Embedded below — or open full-screen [P](#) to interact with it directly.

Mockup for: [admin-admissions-pipeline.html](#) [Full-screen](#)

MathuJ
International School of Oman

Dashboard Students **Admissions** Faculty Reporting [Admissions: My School](#)

Admissions

Keeping our current students for 2026/27, and bringing in the new ones. Combined view for the principal and admissions office.

Re-enrolment is at 88% with 12 days left — 2 points behind last year at the same point. 47 families haven't decided yet. 3 are flagged high-risk on the retention dashboard. The new applicant pipeline is healthy (442 active) but 57 conversion is lagging.
Drafted by MathuJ AI - refreshed today - covers both 3-year timelines.

CURRENTLY ENROLLED
612
Current academic year - all grades

RE-ENROLLED FOR 2026/27
547 (89.4%)
vs 512 pts same time last year - +3.8 pts

NOT YET DECIDED
47
12 to drop to deadline - 3 high-risk

CONFIRMED LEAVING
18
50% graduating 12 - moving 4 - other 3

RE-ENROLMENT — FAMILIES WHO HAVEN'T DECIDED 47 pending - 12 days to deadline - sorted by risk

4 **Khalid Al-Mahdi** - 12Y - 10Y - 3 years at ISD
1. Flagged high-risk on retention dashboard — attendance 4, parent unmet contacts

NO RE-ENROLMENT RISK [Open retention plan](#)

6 **Mays Al-Lawati** - 10Y - 10Y - 4 years at ISD
1. Likely financial blocker — Term 3 installment 22 days overdue - Surplus engaged

DEAR BY TERM [View detail plan](#)

3 **Aisha Al-Bukhari** - 10Y - 10Y - 4 years at ISD - top 5%
1. Family noted at competitor open day - reference letter requested

CONNECTION RISK [Schedule retention call](#)

1 **Rana Al-Said** - 10Y - 10Y - 3 years at ISD
2. Senders wait, home opened - no other negative signals

LIKELY RISK [Send AI nudge](#)

5 **Omar Al-Habib** - 10Y - 10Y - 3 years at ISD
1. Father confirmed at PTC but "has 5 next week" - 11 days ago

FORMAL COMMENT [Send friendly reminder](#)

42 **Other pending families**
No risk signals - routine reminder cycle

ROUTINE [Open all](#)

CONFIRMED LEAVING — AND WHY 18 families - drives next year's planning

GRADUATING **12** 100% leaving — expected - all university destinations confirmed

MOVING COUNTRY **4** 2 to UK, 1 to Canada, 1 to KSA - parent job relocations

SWITCHING SCHOOLS **2** 1 timetable conflict - 1.8 pathway gap — both flagged in exit interview

NEW APPLICANT PIPELINE - 2026/27 142 active - 11 test - funnel conversion at each stage

INQUIRY **142**
47 sources - +18 this week

FOUR REQUESTS **84**
58% of inquiries - +15 this week

ASSESSMENT **61**
70% of tests - 11 completed

INTERVIEW **38**
60% of assessments - 27 waiting

OFFER SENT **31**
80% of interviews - 5 awaiting

ENROLLED **22**
71% of offers - 8 deposits taken

NEEDS YOU THIS WEEK 16% of re-enrolment + new applicants - sorted by impact

3 **High-risk re-enrolments to call**
Khalid, Mays, Aisha — bring any of these needs - +150 30s follow-ups — adding effort this.

42 **routine re-enrolment reminders**
Priority 40 drafted nudges. In the school's inbox 10 days to deadline.

3 **new inquiries unanswered > 48h**
Industry benchmark is 24h. Faster response correlates with 1.5x year-over-year rate.

Open list

T1 assessments overdue (new)

Meet w/ JJ and JJ. All drafted "let's get you scheduled?" follow up meeting

10/1

3 offers awaiting decision (new)

Deadline this week. Nothing drafted

Plan replacement capacity

18 confirmed leaving. 11 need - 82 new assessments to net +65. Currently on track at

Back to all meetings

Meeting meeting library of 0

M

Manhaj

International School of Oman

Dashboard

Students

Faculty

Finance

Reporting

Principal - Dr. Patel

DP

At-risk students & retention

Students drifting on grades, attendance, communications engagement, or fees — flagged early, with a recommended next step for each.

M

7 students flagged this week — 2 high-risk warrant a call before the weekend. Khalil's combined signals (attendance 4-, English grade 4-, parent comms unread) suggest a possible disengagement. Maya's flag is purely financial and easier to resolve.
Drafted by Manhaj AI - last refreshed today - model uses 90-day trailing window

TOTAL FLAGGED

7

of 612 students

HIGH RISK

2

call this week

MEDIUM RISK

3

monitor + message

LOW / INFORMATIONAL

2

keep an eye on

All 7

High 2

Medium 3

Low 2

Financial 2

Academic 3

Engagement 3

Search by student or parent name

HIGH RISK — ACT THIS WEEK

2 students

K

Khalil Al-Mahri

Grade 7B - 2 years at ISO - class teacher Ms. Khadija

Flagged 14 days ago - signals worsening - parent: Salem Al-Mahri

HIGH RISK

82

100

Attendance 95% → 78%

English grade B+ → C- (2 cycles)

Parent unread comms - 4 weeks

Homework completion 92% → 71%

No incidents - no fee issues

M

WHY THIS STUDENT WAS FLAGGED

Three independent signals moved in the same direction over the last 6 weeks. The drop began after Khalil's father missed the Term 2 parent-teacher conference. Khalil's older sister Reema (now at a competitor school) followed a similar pattern in 2024 before leaving. **Recommended:** short personal call from Dr. Patel to the father this week, not an email.

Owner: Dr. Patel

suggested call window Tue-Wed

View Khalil's profile

Draft AI call brief

Schedule call now

M

Maya Al-Lawati

Grade 5A - 4 years at ISO - class teacher Mr. Tariq

Flagged 9 days ago - financial - parent: Sara Al-Lawati

HIGH RISK

71

100

Term 3 installment overdue 32 days

3 reminder emails unanswered

Attendance 98% - stable

Grades stable

No re-enrolment form for 2024/25

M

WHY THIS STUDENT WAS FLAGGED

Academic signals are healthy — this looks purely financial. The unanswered reminders suggest the parent may be uncomfortable raising a payment issue. **Recommended:** a soft, private check-in from the bursar offering an installment-plan conversation — not another reminder email.

Owner: Bursar's office

last contact attempt 3 days ago

View payment history

Draft soft check-in

Assign to bursar

MEDIUM RISK — MONITOR AND MESSAGE

3 students

H

Hassan Al-Said

Grade 3A - 3 years at ISO

Flagged 5 days ago - academic dip

MEDIUM

54

100

Homework completion 90% → 65%

Maths quiz scores down 12%

Attendance 100%

Engagement strong (parent reads everything)

M

WHY THIS STUDENT WAS FLAGGED

Pattern matches "post-illness slump" seen in 12 other G3 students last year — bounces back with a single check-in. **Recommended:** short note from the class teacher to the parent.

Owner: Mr. Khalid

class teacher

View profile

Draft teacher note

A

Aisha Al-Bulushi

Grade 10B - 6 years at ISO

Flagged 2 days ago - competitive admissions signal

MEDIUM

48

100

Reference letter requested (Ms. Khadija)

Family noted at competitor open day (logged by Ms. Salwa, admissions)

Grades excellent - top 5% of cohort

No fee issues

M

WHY THIS STUDENT WAS FLAGGED

Combination of a teacher reference-letter request + the staff-logged observation about the competitor open day is the strongest pre-leave indicator we see for senior years. **Recommended:** a value conversation about ISO's IB diploma pathway and university placement support. Signals are sourced only from connected school data and observations submitted by your team — never from social-media scraping.

Owner: Dr. Patel

sensitive

View profile

Schedule retention conversation

Z

Zayn Al-Maamari

Grade 6C - 4 years at ISO

Flagged 11 days ago - behaviour shift

MEDIUM

42

100

3 behaviour notes (was 0 last term)

Grades stable

Attendance 96%

Parent stopped opening weekly digests

M

WHY THIS STUDENT WAS FLAGGED

Behaviour change at age 11-12 often precedes a request to move schools "for a fresh start". Currently mild. **Recommended:** counsellor check-in with Zayn first, parent follow-up second.

Owner: Ms. Reem

counsellor

View profile

Assign to counsellor

LOW / INFORMATIONAL

2 students - no action required, just visibility

N

Nasser Al-Rashdi

Grade 6A - attendance one outlier day

Flagged 3 days ago - single missed day after pattern of perfect attendance

LOW

22

100

No owner assigned - low priority

Dismiss flag

S

Sara Al-Said

Grade 7C - re-enrolment form not yet returned

Flagged 1 day ago - 30 days from deadline - no other concerns

LOW

18

100

Likely fine — included in next re-enrolment batch reminder

Dismiss flag

At-Risk / Retention Dashboard

At-Risk / Retention Dashboard

Students at risk of dropping out of school, or who are struggling academically, or who are struggling financially, or who are struggling with attendance, or who are struggling with engagement, or who are struggling with retention.

Visual summary Backed & data The mockup

[Open mockup full screen](#)

1. VISUAL SUMMARY

What this screen is

Students at risk of dropping out of school, or who are struggling academically, or who are struggling financially, or who are struggling with attendance, or who are struggling with engagement, or who are struggling with retention.

WHO IT'S FOR

Principal, head of school, head of pastoral care.

WORKFLOW OF SUPPORTS

Catches families before they leave. Differentiates available attrition (see can fix this) from unavoidable (family relocating).

KEY FEATURES

- At snapshot framing the week's risk picture
- Stats: total flagged, high / medium / low risk counts
- Filter chips by risk level and by cause (Financial / Academic / Engagement)
- Rich student cards with signal chips, sparkline trends, AI reasoning, recommended action
- Clear owner assignment (Dr. Patel / Bunar / Counsellor)
- Action buttons specific to the case (Schedule call, Draft AI brief, Soft check-in)
- Signals sourced from connected school data and staff-submitted observations only — no social-media scraping

2. BACKEND & DATA

What it takes to build this for real

Data inputs, backend services, AI rules, and privacy considerations.

INPUT DATA REQUIRED

- Per-student attendance trend (last 30 days)
- Per-student grade trend
- Per-student parent-comm engagement (align opens, message reads)
- Per-family financial status (invoice history, age of debt)
- Per-student behaviour notes (positive + incidents)
- Staff-submitted observations (e.g. "family at competitor open day" flagged by admin)
- Historical retention outcomes (for the ML model to learn from)

BACKEND BUILT REQUIRED

- Daily ETL aggregating per-student signal trends
- Risk model: starts as rule-based heuristics (3+ negative signals + comm unread = flag); evolves to weighted regression once historical outcomes accumulate
- Reasoning generation via Claude (explains why each student was flagged, what the pattern matched)
- Action assignment workflow (suggested owner, ability to reassign)
- Integration with messaging (SMS, TTS) for one-click drafts
- Privacy: strict role-gating — only principal + designated admin roles see this view

AI ROLE

Heavy pattern detection across signals, generating per-student reasoning, recommending action channel and owner.

PRIVACY & ACCESS

Admin role only. Flagged students never told they are flagged. AI never accesses private parent messages or social media — only school-connected data + staff observations.

3. THE MOCKUP

Live preview

Embedded below — or open full-screen [P](#) to interact with it directly.

Mockup for: [admin-retention-at-risk.html](#) [Full screen](#)

Marble | International School of Oman | Dashboard | **Students** | Faculty | Finance | Reporting | [Principal: Dr. Patel](#)

At-risk students & retention

Students drifting on grades, attendance, communications engagement, or fees — flagged early, with a recommended next step for each.

7 students flagged this week — 2 high-risk warrant a call before the weekend. Khalil's combined signals (attendance ↓, English grade ↓, parent comm unread) suggest a possible disengagement. Mayad's flag is purely financial and easier to resolve. Drafted by Marissa AI. Last refreshed today. Model uses 30-day rolling window.

TOTAL FLAGGED
7
of 612 students

HIGH RISK
2
call this week

MEDIUM RISK
3
monitor + message

LOW / INFORMATIONAL
2
keep an eye on

ALL | High | Medium | Low | Financial | Academic | Engagement

Search by student or parent name

HIGH RISK — ACT THIS WEEK

2 students

Khalil Al-Mahdi | Grade 10 | 2 years at ISS | class teacher Mr. Khalifa | **High Risk** | **\$2,100**

Flagged 14 days ago | signals worsening: parent: Salim Al-Mahdi

Attendance 85% < 90% | English grade B+ < B (20 updated) | Parent unread notices: 4 weeks | Homework completion 90% < 95%

No incidents in the last 30 days

Why this student was flagged
Three independent signals moved in the same direction over the last 6 weeks. The drop began after Khalil's father missed the Term 2 parent-teacher conference. Khalil's older sister Reema (now at a competitor school) followed a similar pattern in 2024 before leaving. **Recommended:** short personal call from Dr. Patel to the father this week, not an email.

Owner: Dr. Patel | suggested call window: Tue-Thu | [View Khalil's profile](#) | [Draft AI call brief](#) | [Schedule call now](#)

Mayad Al-Lawati | Grade 10 | 4 years at ISS | class teacher Mr. Tariq | **Medium Risk** | **71**

Flagged 8 days ago | Financial | parent: Tarek Al-Lawati

Term 2 installment overdue 30 days | 3 reminder emails unanswered | Attendance 95%: stable | Grades stable

No incidents in the last 30 days

Why this student was flagged
Academic signals are healthy — this looks purely financial. The unanswered reminders suggest the parent may be uncomfortable raising a payment issue. **Recommended:** a soft, private check-in from the bursar offering an installment plan conversation — not another reminder email.

Owner: Bunar's office | last contact attempt: 3 days ago | [View payment history](#) | [Draft soft check-in](#) | [Assign to Bunar](#)

MEDIUM RISK — MONITOR AND MESSAGE

3 students

Hassan Al-Said | Grade 10 | 2 years at ISS | **Medium** | **\$4,100**

Flagged 5 days ago | Academic only

Homework completion 80% < 85% | Maths quiz scores down 15% | Attendance 100% | Engagement strong: parent reads everything

Why this student was flagged
Pattern matches "good stress slump" seen in 12 other SS students last year — sources back with a single check-in. **Recommended:** short note from the class teacher to the parent.

Owner: Mr. Khalifa (class teacher) | [View profile](#) | [Draft teacher note](#)



Concern/incident records (safeguarding): Phase 1 captures concern notes through `behaviour_notes` + at-risk `risk_flags` ; the full safeguarding case-management workflow is Phase 2. The demo log is shown for context:

Safeguarding & incident log

Confidential record of safeguarding concerns, medical events, accidents, behaviour, and bullying. Access is role-restricted and every view is audited.



You're viewing this as **Designated Safeguarding Lead** — you see all categories. Non-DSL staff see only the categories assigned to their role, and child-protection records are restricted to you and the Principal. Every view is logged.



4 open cases - 1 statutory deadline this week. The Year 7 bullying investigation is at day 4 of the 10-day MoE-required cycle. A pattern is forming in the playground at lunchtime — 3 incidents in 7 days, all at the south climbing frame.

Drafted by Manhaj AI - summaries only - does not see redacted records

LOGGED THIS WEEK

11

vs avg 14 - slightly below trend

OPEN CASES

4

1 statutory - 3 routine

CLOSED THIS WEEK

8

Avg time-to-close: 2.4 days

STATUTORY DEADLINE

1

Y7 bullying report — 6 days left

OPEN CASES REQUIRING ATTENTION

[+ Log new incident](#)

STATUTORY - CHILD-PROTECTION - DSL + PRINCIPAL ONLY

Opened 28 May - day 4 of 10



Case 2026-CP-002 — referred to Ministry of Social Development

Category **Child protection** - Assigned to **you (DSL)** - **Principal** notified

Record contents are restricted. Open the case file to view — your access will be logged.

ACTIVE INVESTIGATION

Next action: external agency call by Mon 2 Jun

[Open case](#)

STATUTORY DEADLINE - BULLYING - YEAR 7

Opened 27 May - day 4 of 10



Bullying complaint — Year 7B - 3 students involved

Category **Bullying** - Assigned to **Ms. Reem (DSL)** + **Ms. Khadija (class teacher)**

Verbal complaint from one parent following a series of WhatsApp messages between three Y7B students. Mediation session held Wed 29 May. Follow-up parent meeting scheduled Mon 2 Jun. Behaviour plan drafted for all three.

INVESTIGATING

MoE return due Tue 9 Jun - 6 days

[Draft MoE return](#)[Open case](#)

MEDICAL - ALLERGY EXPOSURE - MONITORING

Opened 29 May - day 2



Allergy reaction monitoring — Layla Al-Habsi - Grade 5B

Category **Medical - allergy** - Reported by **school nurse** - Parent notified **same day**

Mild reaction to peanut residue at lunch on 29 May — antihistamine administered, no EpiPen required. Parent informed within 30 minutes. Lunch supplier briefed; review of nut-free zone signage initiated. Continue 5-day monitoring before closing.

MONITORING

Close after 3 more days - parent updates due

[Open case](#)

ACCIDENT - MINOR INJURY

Opened 28 May - day 3



Head bump — Hassan Al-Said - Grade 3A

Category **Accident - minor** - Logged by **playground supervisor** - Parent collected

Fell from the south climbing frame at lunch. Bump on forehead, ice applied, no concussion symptoms. Parent collected at 1:15 PM. No medical attention sought. **Note: 3rd playground incident at this location in 7 days.**

ACTION PLAN

Facilities review of south climbing frame

[Add to facilities log](#)[Close case](#)

PATTERNS THE AI SURFACED

Look at these — they may already be a problem



South climbing frame — 3 incidents in 7 days

All minor, all lunchtime. **Suggest:** facilities check + supervisor reassignment.



Tuesday lunch — 60% higher incident rate

Across all categories. Coincides with reduced supervisor coverage on Tuesdays. **Suggest:** rota adjustment.



Year 7 — 4 behaviour notes in 2 weeks

Below the bullying-investigation threshold but worth observing. **Suggest:** counsellor walk-through.

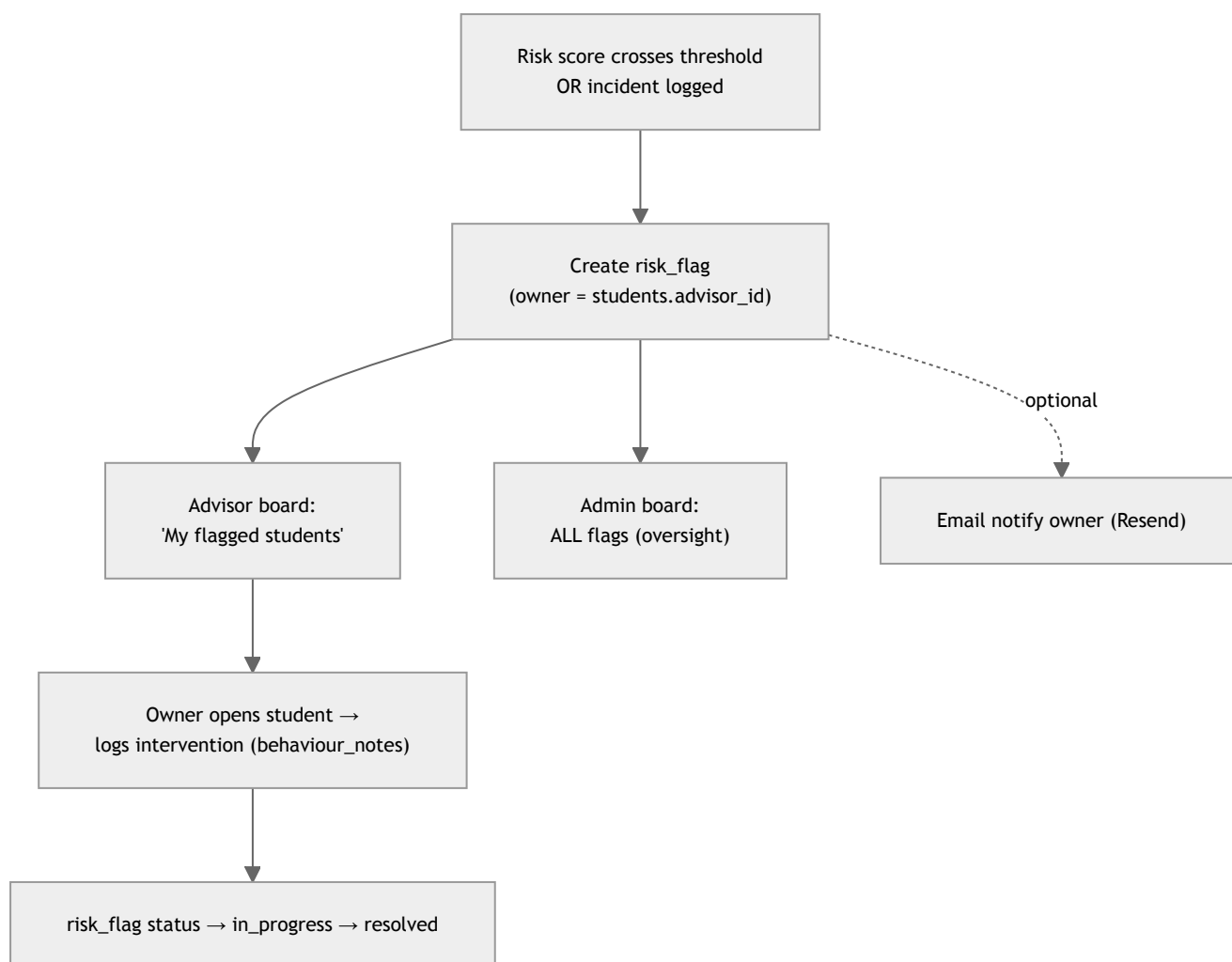
RECENTLY LOGGED - LAST 14 DAYS

11 entries - full log →

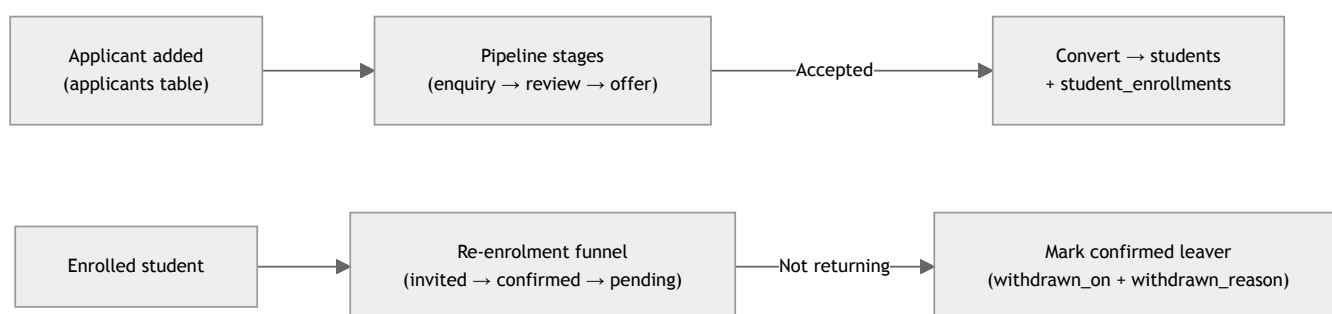
29 May 12:45	MEDICAL	Layla Al-Habsi (G5) allergy reaction monitoring opened Reported by school nurse - parent notified same day	OPEN
28 May 13:10	ACCIDENT	Hassan Al-Said (G3) playground head bump South climbing frame - parent collected	OPEN
28 May 09:30	SAFEGUARDING	Case 2026-CP-002 opened Details restricted	OPEN
27 May 14:55	BULLYING	Year 7B - 3 students - WhatsApp messages — investigation opened Parent complaint received 26 May	OPEN
26 May 11:30	MEDICAL	Maya Al-Lawati (G5) menstrual care provided Clinic logged - routine	CLOSED
23 May 16:40	BEHAVIOUR	Zayn Al-Maamari (G6) classroom disruption — counsellor referral Pattern noted - linked to retention dashboard	OPEN
22 May 10:15	ACCIDENT	Salim Al-Lawati (G1) scraped knee South climbing frame - parent informed	CLOSED
21 May 12:55	PROCESS	Lost child located — Yusuf Al-Habsi (K02) found in library after 4 mins Procedure review logged - no harm	CLOSED

Who uses it & permissions. Admin (full, sees ALL flags). Advisor sees **only their owned students** (RLS scoped to `risk_flags.owner_id`). The homeroom teacher (Teacher app) is the other possible owner — see open question.

Workflow — at-risk ownership routing (the important mechanic).



Workflow — admissions / enrolment lifecycle.



Backend & data linkage.

- **At-risk:** the grading mechanism is kept as-is. A job/trigger creates a `risk_flags` (NEW) row owned by `students.advisor_id` (NEW column). RLS scopes advisors to *their* flags; admins see all. Logging an intervention uses the existing `behaviour_notes` pattern and updates the flag status. This is the one mechanic that reaches into the **teacher app** (homeroom teacher as owner) — confirm owner precedence. *Terminology note:* the at-risk "**advisor**" here may be the same person the Student doc calls the "**counsellor**" (university/guidance) — one role, two names across archetypes.
- **Re-enrolment funnel** reads enrolment status; "show more" expands the not-enrolled list. Marking a **confirmed leaver** sets `students.withdrawn_on` + the NEW `students.withdrawn_reason`.
- **Applicants** live in the NEW `applicants` table (stage, contact, target grade). "Convert" is a server action that creates the `students` + `student_enrollments` rows and closes the applicant — the applicant→student bridge.

Data tables.

NEW `risk_flags` — routed at-risk flags. `student` and `owner (advisor)` show names as display stand-ins for the uuid FK.

student	owner (advisor)	severity	reason	status	created_at
Khalil Al-Mahri	N. Al-Kindi	high	attendance ↓ + 2 missed quizzes	in_progress	2026-06-19T06:00:00Z
Salim Al-Lawati	N. Al-Kindi	medium	comms-engagement drop	open	2026-06-21T05:00:00Z

NEW `applicants` — admissions pipeline:

name	target_grade	contact (email)	stage	source	created_at
Maryam Al-Busaidi	6	parent.busaidi@mail.om	offer	referral	2026-06-10T09:00:00Z
Yusuf Al-Harthy	9	parent.harthy@mail.om	review	website	2026-06-18T14:30:00Z

`students` (existing + **NEW** `advisor_id`, `withdrawn_reason`) — leaver record. The `advisor_id` value shows an advisor name as a display stand-in for the uuid FK.

full_name_en	advisor_id NEW	withdrawn_on	withdrawn_reason NEW
Layla Al-Habsi (G10)	N. Al-Kindi	—	—
Omar Al-Rawahi	N. Al-Kindi	2026-06-15	family_relocating

Deferred to P2/P3. Full safeguarding case management (the `admin-safeguarding-log` workflow) → Phase 2. *Why (workflow terms):* P1 already records the *signal* (a concern note / a risk flag) and routes it to an owner; the additional case lifecycle (open→investigate→escalate→close with statutory fields) is a distinct workflow with its own audit requirements. *Clean-linkage:* build it as a `safeguarding_cases` table that **links** to existing `behaviour_notes` / `risk_flags` rows (1 case ← many notes), so nothing P1 writes needs migrating.

Open questions.

- At-risk **owner** (D8): advisor only, homeroom teacher only, or both — and routing precedence. Shapes the teacher app too.
- Leaver-reason vocabulary — confirm the school's categories (feeds MoE reporting in A8).

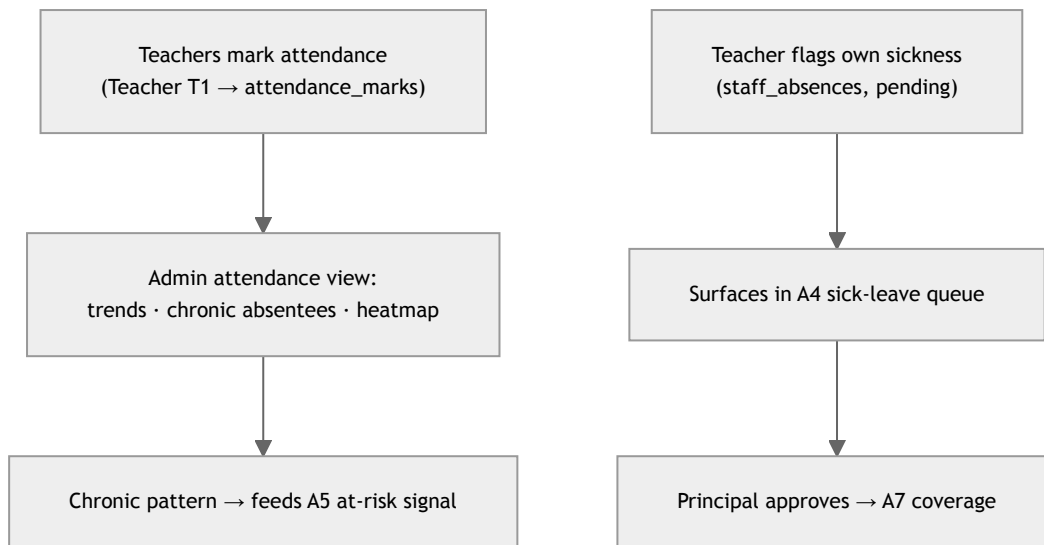
6. A6 — Attendance

What it is. The admin's attendance lens: trend/heatmap/chronic-absentee views over the marks teachers write each period, plus the **teacher-sickness intake** wiring that flows into the Faculty approval queue (A4 / D10). The admin does not normally *mark* attendance — that's the teacher (Teacher T1) — but can do an end-of-day backfill on a teacher's behalf, and reads the aggregate here.

Visual. (No dedicated mockup; the admin attendance view is composed of trend/chronic-absentee panels over `attendance_marks`. The sick-leave intake half is visualised in §4/§7.)

Who uses it & permissions. Admin/Principal (view + approve sick-leave). Advisor (their cohort only). Teachers mark + flag own sickness in the Teacher app.

Workflow.



Backend & data linkage.

- The trend/chronic-absentee panels are **read-only aggregations** over `attendance_marks` (existing: `student × bell_period × marked_on`, `status` enum). No new write path on the admin side.
- Chronic-absence patterns are one input to the A5 at-risk signal (which generates `risk_flags`).
- The **sick-leave intake** half shares `staff_absences` with A4 — the teacher writes the `pending` row; the admin acts on it in the Faculty queue. Surfaced in both tabs, single source of truth.
- Downstream, the daily roll-up also feeds the **Parent weekly digest (P1)**.

Data tables.

`attendance_marks` (existing) — read by the admin aggregation (G5B, Period 2). Names in the `student` / `marked_by_teacher_id` columns are display stand-ins for the uuid FK.

student	marked_on	bell_period	status	reason	marked_by_teacher_id
Khalil Al-Mahri	2026-05-31	P2	absent	family_informed	Ms Swart
Maya Al-Lawati	2026-05-31	P2	late	bus_late	Ms Swart
Sara Al-Mughairi	2026-05-31	P2	present	—	Ms Swart

(Sick-leave `staff_absences` rows shown in §4.)

Deferred to P2/P3. Rich attendance analytics (predictive chronic-absence modelling) → Phase 2; P1 ships the descriptive views. **Reception / gate pickup feed** (`admin-reception-pickup`) → Phase 2 — it needs a gate check-in data source not present in P1.

Open questions.

- Attendance cadence (per-period vs once-daily homeroom) — same school question as Teacher T1; affects the volume the admin aggregates.

7. A7 — Schedule + auto-scheduling

What it is. Build and maintain the weekly timetable, find daily substitutes, and **auto-generate** a timetable from the teacher-load matrix. The principal can type **plain-English constraints** ("keep Mr Gericke's Tuesday afternoons free") and resolve conflicts/gaps. The killer Phase-1 feature is the **daily substitute finder** — when an absence is approved (A4), it ranks qualified subs and assigns one, which fires the Teacher T6 handoff. This is the heaviest engineering item in Phase 1.

Visual.

M

Manhaj

International School of Oman

Dashboard

Students

Faculty

Schedule

Reporting

Operations - Mr. Sami

SA

Today's schedule

Sunday 31 May 2026 - check coverage, find substitutes, see the full school day.

Today

This week

Master timetable

Cover history

M

1 teacher absent today, 4 periods needing cover — all 4 now assigned. Ms. Fatima covers Mr. Tariq's whole day - she's accepted and the handoff sheet is in her inbox. No other gaps. School is fully staffed.
Manhaj auto-matched substitutes last night - Mr. Sami confirmed at 22:14

CLASSES TODAY

47

across 32 sections

TEACHERS ABSENT

1

Mr. Tariq - sick leave

COVER ASSIGNED

4 / 4

All periods covered

OPEN GAPS

0

School fully staffed

TODAY'S SUBSTITUTIONS

1 absence - 4 periods - all assigned

TQ

Mr. Tariq Al-Bahri - Maths

Out today - sick leave reported 21:30 last night - returns tomorrow

4 teaching periods

✓ FULLY COVERED

08:00

P1 - 45 min

G5B Maths - Room 12 - 22 students

KHALID NEEDS CATCH-UP

LAYLA ALLERGY

Ms. Fatima Al-Hinal

Match 96 - accepted 22:18

✓

WHY MS. FATIMA - RANKED CANDIDATES

Change pick

FA

Ms. Fatima Al-Hinal

G5 Maths qualified - free - low load - same dept

96

KM

Mr. Khalid

Head of Maths - free P1 only - high load

74

QA

Mr. Omar

Maths - not free at 08:00 - already teaching G7B

—

09:00

P2 - 45 min

Free period - no class

No cover needed

09:45

P3 - 45 min

G3A Maths - Room 8 - 19 students

HASSAN - POST-INJURY WATCH

Ms. Fatima Al-Hinal

Match 92 - same sub all-day - accepted 22:18

✓

10:45

P4 - 45 min

G5B Maths - Room 12 - same class as P1

Ms. Fatima Al-Hinal

Match 97 - continuity with P1 - accepted 22:18

✓

12:00

Lunch

Lunch - staff room

—

12:45

P5 - 45 min

G6C Maths - Room 14 - 24 students

ZHAO LI (EAL)

Ms. Fatima Al-Hinal

Match 89 - whole-day cover - handoff sent

✓

13:30

P6 - 45 min

Free period

No cover needed

Handoff sheet - sent to Ms. Fatima at 22:14 last night - she's read it at 06:42 today

View handoff sheet

Message Ms. Fatima

Mark Mr. Tariq returned

MASTER TIMETABLE - THIS WEEK

G5B - Sun - Thu - class-view default - switch to teacher-view →

	SUN 31	MON 1	TUE 2	WED 3	THU 4
P1 08:00	Maths Ms. Fatima (sub)	Arabic Ms. Maryam	English Ms. Sara	Field trip Bait Al Zubair	Maths Mr. Tariq
P2 09:00	free	Science Mr. Ahmed	Maths Mr. Tariq	Field trip	English Ms. Sara
P3 09:45	Maths Ms. Fatima (sub)	Maths Mr. Tariq	PE Coach Issa	Field trip	Science Mr. Ahmed
P4 10:45	Maths Ms. Fatima (sub)	English Ms. Sara	English Ms. Sara	Field trip	Arabic Ms. Maryam
Lunch 12:00	Lunch - 45 min				
P5 12:45	Maths G6C Ms. Fatima (sub)	free	Arabic Ms. Maryam	Field trip	Maths - TEST Mr. Tariq
P6 13:30	free	PE Coach Issa	free	Field trip	English review Ms. Sara

Plan next week's timetable

Bring up the constraint solver to draft week of Sun 7 June - 1 known absence (Mr. Khalid - conference) - auto-suggestions ready.

Plan next week →

[Visual summary](#) [Backend & data](#) [The workflow](#)

Open market full screen 2

© 2000 Blackwell Science Ltd, *Journal of Internal Medicine* 247: 399–407

Today's schedule view with coverage status, AI-ranked substitute matching for absences, and a read-only master timetable. The killer feature is daily substitute finding — not the once-a-year timetable build.

www.elsevier.com/locate/jmb

Operations head, principal, or anyone managing daily school affairs.

with the same level of confidence.

When a teacher calls in sick, automatically finds qualified substitutes, ranks by fit (subject competence + availability + load + dept + recency), assigns, and triggers the handoff sheet (TS).

NEW FEATURES

- View tabs (Today / This week / Weather timetable / Cover history)
- Day state classes, teachers absent, cover assigned, open gaps
- Absent teacher block showing each affected period
- Per-period candidate ranking with match score (subject + free + load + dept)
- One-click assign = triggers handoff sheet to substitute (TE)
- Same substitute all-day preference reflected in scoring
- Master timetable grid (read-only) with SUB-tagged covered cells
- Plan next week forward navigation feed to constraint solver

D. J. Henderson and A. H. Hurrell

Data inputs, backend services, AI rule, and privacy considerations.

INPUT DATA REQUIRED

- Master timetable (per teacher / per class / per period)
- Teacher subject competencies (who's qualified to teach what)
- Teacher availability today (free periods, plus absence flags)
- This-week and lifetime substitute-load per teacher (fairness)
- Department assignments
- Recent sub-assignments (to avoid over-using one person)

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- Teacher subject competency table
- Daily availability state (five periods derived from timetable, modified by absences)
- Substitute matching SQL with ranked scoring (subject hard-filter, then weighted soft factors)
- One-click assignment workflow that pushes to 76 handoff generator
- Same-substitute-all-day preference logic in the tanker
- Master timetable: stored relational per period + class + room + teacher)
- Plan next needs: CP-SAT solver (0s to seconds) – separate lib build for the newer use cases

1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 2676, 2677, 2678, 2679, 26

Light for the daily matcher (deterministic ranking). Heavy for the annual timetable solver (CP-SAT constraint optimisation — separate feature).

PRIVATE & ACCESS

Admin rule. Teacher absence reasons may be restricted (i.e., personal don't need to be shared with all staff)

6. Feed schedule

Embedded below — or open full-screen [P](#) to interact with it directly.

Maths@M | **Maths@M** | International School of Oman | Dashboard | Students | Faculty | Schedule | Reporting | Operations: Mr. Tawfik

Today's schedule

Sunday 31 May 2020 - check coverage, find substitutes, view the full school day.

Today | This week | Master timetable | Cover history

W 1 teacher absent today, 4 periods needing cover – all 4 now assigned. Ms. Fatima covers Mr. Tariq's whole day - she's accepted and the handoff sheet is in her inbox. No other gaps. School is fully staffed.
Missing note-matched substitute last night. Mr. Sam confirmed at 20:16.

Covers Today	Teachers Absent	Covers Assigned	Open Gaps
47 covered 10 sections	1 Mr. Tariq - sick leave	4 / 4 All periods covered	0 School fully staffed

TODAY'S SUBSTITUTIONS

1 absence - 4 periods of assigned

TS **Mr. Tariq Al-Sabhi - Maths**
Out today - which leaves reported 21:00 last night - returns tomorrow

4 teaching periods
→ **75%+ coverage**

G5B Maths	Room 10 - 22 students	Handoff: missing lesson plan	Ms. Fatima Al-Husni	March 02 - accepted 22:16
06:00 P1 - 45 min Topic: Equivalent Fractions Intro - Lesson plan + Materials in room	MS Ms. Fatima Al-Husni No lesson available? Hand off next same day	36	MS Ms. Al-Hadi Lesson plan ready - Handoff ready	34

WHY MS. FATIMA - RANKED CANDIDATES Change pick

MS	Ms. Fatima Al-Husni	Ms. Al-Hadi	Ms. Omar
MS Ms. Fatima Al-Husni No lesson available? Hand off next same day	MS Ms. Al-Hadi Lesson plan ready - Handoff ready	MS Ms. Omar Lesson plan ready - Handoff ready	MS Ms. Tariq Lesson plan ready - Handoff ready

09:00 P2 - 45 min Free period - no class	No cover needed
09:45 P3 - 45 min G5A Maths Room 8 - 19 students Topic: Multiplication tables review - routine lesson	MS Ms. Fatima Al-Husni March 02 - same roll as all-day - accepted 22:16
10:45 P4 - 45 min G5B Maths Room 10 - same class as P1 Topic: Exponents practice (continues P1) - self-led	MS Ms. Fatima Al-Husni March 02 - continuity with P1 - accepted 22:16
12:00 Lunch Lunch - staff meals	---
12:45 P5 - 45 min G5C Maths Room 14 - 24 students Topic: Decimal fractions intro - slides ready on board	MS Ms. Fatima Al-Husni March 02 - already off cover - handoff sent
13:30 P6 - 45 min Free period	No cover needed

Handoff sheet - sent to Ms. Fatima at 22:16 last night - she's email'd it at 00:42 today

View handoff sheet Message Ms. Fatima Mark Mr. Tariq returned

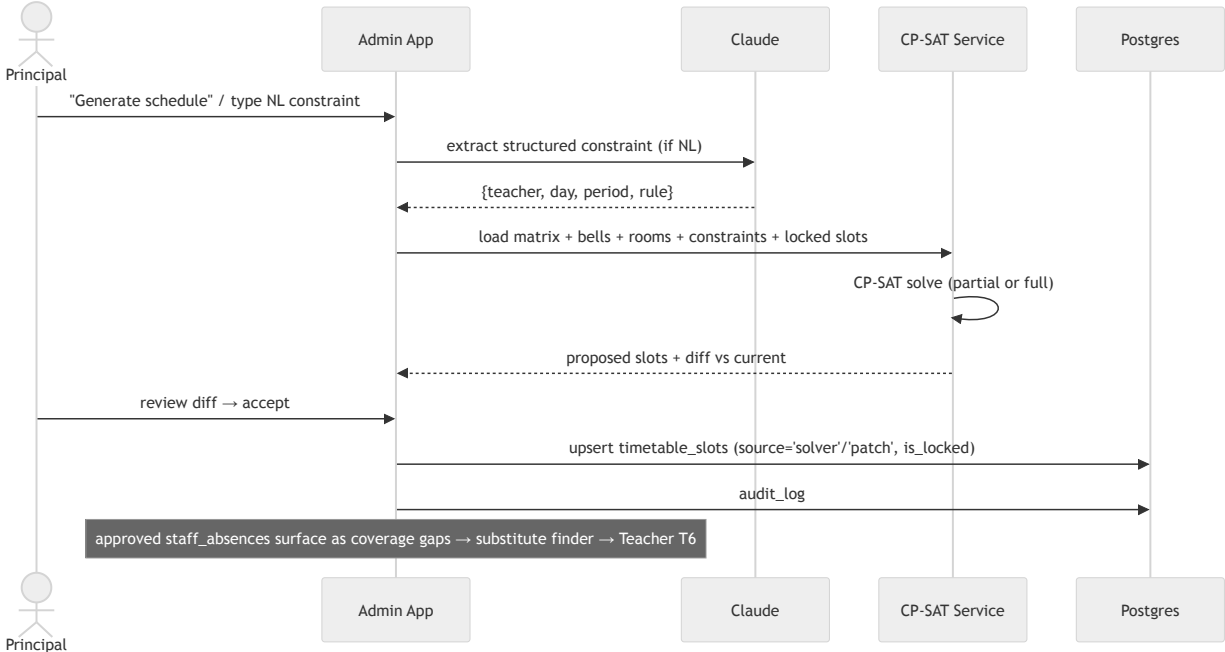
MASTER TIMETABLE - THIS WEEK

G5B - Sun - The class view default - switch to teacher view

#11 2024-01-01	Maths Mr. Tariq (sub)	Maths Mr. Tariq	English Mr. Tariq	Handing Mr. Tariq	Maths Mr. Tariq
#12 2024-01-02	Maths Mr. Tariq	Maths Mr. Tariq	English Mr. Tariq	Handing Mr. Tariq	Maths Mr. Tariq
#13 2024-01-03	Maths Mr. Tariq	Maths Mr. Tariq	English Mr. Tariq	Handing Mr. Tariq	Maths Mr. Tariq

Who uses it & permissions. Admin/Principal only. Writes are guarded; human-edited slots are `is_locked` to protect them from the solver.

Workflow.



Backend & data linkage.

- Inputs the solver reads:** `teacher_section_subject` (the load matrix — who teaches what, how many periods), `bell_periods` (when), `rooms` (where), `sections`, `teacher_contracts` (caps), and any **locked** existing `timetable_slots`.
- NL constraints:** Claude parses free text into a structured constraint; the solver does a **partial re-solve** touching only affected slots, returns a diff for human approval. One `ai_usage_ledger` row per parse.
- Writes:** approved results upsert `timetable_slots` with `source = solver | human | patch` and `is_locked` to protect human decisions.
- Substitute finder:** approved `staff_absences` (A4) surface as coverage gaps; the finder ranks qualified subs (subject competence + availability + load via `v_teacher_load` + recency), the admin assigns `staff_absences.sub_teacher_id`, and that assignment **fires the Teacher T6 substitute handoff** (Teacher doc §4).
- The solver runs as a **separate FastAPI service** (OR-Tools CP-SAT, Fly.io/Railway) called from a server action — *not* in the Next.js process.

Data tables.

`timetable_slots` (existing) — after an accepted solve:

section	subject	teacher	room	bell_period	source	is_locked
G5B	Maths	Mr Tariq Al-Bahri	R12	Sun-P1	solver	false
G9A	Maths	Ms S. Swart	R12	Sun-P2	human	true
G5B	Maths	Ms Fatima Al-Hinai (sub)	R12	Sun-P1	patch	false

`ai_usage_ledger` (existing) — one row per NL-constraint parse:

request_kind	model	input_tokens	output_tokens	cost_usd	is_background
schedule_constraint	claude-sonnet-4-5	640	90	0.0042	false

Deferred to P2/P3. None — A7 is Phase 1. (Room-utilisation analytics and multi-week pattern optimisation are P2 niceties layered on the same `timetable_slots`.)

Open questions.

- Substitute ranking weights (subject vs availability vs recency) — confirm with the school's actual cover policy (sales to source).
- Does the principal want full auto-generate from scratch in P1, or auto-generate-then-tune only? (Affects solver scope.)

8. A8 — Reports

What it is. Two halves. (1) The **parent-comms pipeline** (pick template → draft → approve → send → track) — buildable now, email-only via Resend. (2) The **regulator-mandated reports** (Oman MoE / KHDA / ADEK / SPEA) — **blocked** until we source the Ministry requirements (templates, mandated content, deadlines). Phase 1 also adds **report storage + a tracking table with retention** (X16) and a **read-only assessment/exam oversight view** (X13).

Visual.

M

Manhaj

International School of Oman

DashboardStudentsFacultyFinanceReporting

Principal · Dr. PatelDP

Regulator reporting

One-click exports in the format each regulator expects — pre-filled from your live data, audit-ready, signed PDF on submit.

Oman MoE

KHDA

ADEK

SPEA

Showing reports for International School of Oman · all data refreshed today

M

3 reports due in the next 30 days, 1 needs attention now. Term 3 attendance summary (due 30 June) is 92% ready — staffing roster needs 2 missing licence numbers.
Drafted by Manhaj AI · refreshed today at 8:30 AM

UPCOMING SUBMISSIONS

Next 90 days · Oman MoE

Q2 incident & safeguarding return

Form MoE-SG-04 · covers 1 Mar — 31 May 2026 · 4 incidents logged this quarter

DUE IN 6 DAYS

Review draft

Generate & submit

Term 3 attendance summary

Form MoE-ATT-T3 · grades KG1 — G12 · pre-filled from daily attendance data

DUE 30 JUNE

Review draft

Generate

Annual staffing & licensing roster

Form MoE-STF-AN · 69 staff · ⚠ 2 missing licence numbers (Ms. Reem · Mr. Hassan) — click to fix

DUE 15 JULY

Fix & preview

Generate

GENERATE A REPORT

Pre-formatted for the selected regulator

Enrolment by grade

Live headcount per grade with bilingual labels and gender breakdown.
Last gen 14 May

PDF · CSV

Attendance summary

Daily, weekly, or term - per student or aggregated · regulator-required cuts.
Last gen 2 May

PDF · XLSX

Staff roster & licences

Teaching staff with licence numbers, expiry dates, qualifications, contract type.
Last gen 28 Apr

PDF · XLSX

Demographic breakdown

Nationality, language, special-needs flags · in MoE-standard categories.
Last gen 14 Apr

PDF · CSV

Safeguarding & incidents

Logged incidents with response actions · redacted for student privacy.
Last gen 30 Apr

PDF

Curriculum alignment

Subjects taught vs MoE curriculum framework · gaps highlighted.
Last gen 28 Mar

PDF

RECENT SUBMISSIONS

View full history →

REPORT	PERIOD	SUBMITTED	BY	STATUS	
Term 2 attendance summary	1 Jan — 28 Mar 2026	30 Apr 2026	Dr. Patel	ACCEPTED	Download ↓
Q1 incident & safeguarding return	1 Jan — 31 Mar 2026	15 Apr 2026	Ms. Salwa	ACCEPTED	Download ↓
Staffing change notice — new hire	15 Apr 2026	16 Apr 2026	HR	ACCEPTED	Download ↓
Term 1 attendance summary	1 Sep — 19 Dec 2025	10 Jan 2026	Dr. Patel	ACCEPTED	Download ↓
Annual enrolment declaration	2025/26 academic year	5 Oct 2025	Dr. Patel	ACCEPTED	Download ↓

COMPLIANCE SNAPSHOT

Live · refreshed daily

CURRICULUM ALIGNMENT

● Aligned

All grades · last audit 28 Mar

TEACHER LICENSING

● 2 expiring soon

Both renewals due before 30 Jun

SAFEGUARDING TRAINING

● 100% complete

All 69 staff certified this year

HEALTH & SAFETY AUDIT

● Passed

Last inspection 12 Feb · next due Sep 2026

by [David Williams](#)

Regulator Reporting Hub

One-click exports in the format each regulator expects (Ofsted, KHDA, ADCK, SPEA). Pre-filled from live school data, audit-ready.

Visual summary **Backend & data** The workup[Open workup full-screen](#)

1 - VISUAL SUMMARY

What this screen is

One-click exports of statutory reports in the format each regulator expects (Ofsted, KHDA, ADCK, SPEA). Pre-filled from live school data, audit-ready.

WHO IT'S FOR

Principal or designated compliance officer responsible for regulator submissions.

WORKFLOW IT SUPPORTS

Replaces weeks of manual data assembly with one-click generation. Compliance snapshot shows status at a glance.

KEY FEATURES

- Regulator tab switcher (Ofsted / KHDA / ADCK / SPEA) so one screen serves any GCC school
- Upcoming submissions cards with urgency colour-coding and deadline countdown
- 6 quick-export cards: enrolment, attendance, staff licensing, demographics, safeguarding, curriculum
- Recent submissions table with download links
- Compliance snapshot (curriculum alignment, teacher licensing, safeguarding training, iHuma3)
- Inline "fix missing data" links (e.g. "2 missing licence numbers — click to fix")

2 - BACKEND & DATA

What it takes to build this for real

Data inputs, backend services, AI role, and privacy considerations.

INPUT DATA REQUIRED

- Per-regulator schema definitions (acquired from regulator publications)
- Live school data: enrolment, attendance aggregates, staff records, incidents, financial summaries
- Audit trail metadata: who submitted, when, status
- Compliance state per category (refreshed daily)

BACKEND BUILD REQUIRED

- Per-regulator template engine (PDF / XLSX / CSV generators with regulator-specific layouts)
- Schema validation against regulator spec before submission
- Generated-document storage with retention per regulator policy
- Submission state machine (Draft → Submitted → Accepted)
- Where regulator APIs exist (rare): direct submission integration; otherwise manual upload
- Per-regulator config: which fields are required, what cadence

AI ROLE

Medium: AI checks generated reports for obvious errors before submission (missing data, format issues). Does not write the substantive report — that's pure data extraction.

PRIVACY & ACCESS

Internal school data only; no parent or student PII in regulator reports beyond what the regulator requires.

3 - THE WORKUP

Live preview

Embedded below — or open full-screen [P](#) to interact with it directly.

Workup the **school-regulator-reporting** tool Full-screen [P](#)



International School of Oman

Dashboard

Students

Faculty

Finance

Reporting

Principal: Dr. Patel

+

Regulator reporting

One-click exports in the format each regulator expects — pre-filled from your live data, audit-ready, signed PDF on submit.

Ofsted

KHDA

ADCK

SPEA

Showing reports for International School of Oman — all data refreshed today

3 reports due in the next 30 days, 1 needs attention now. Term 3 attendance summary (due 30 June) is 92% ready — staffing roster needs 2 missing licence numbers.

Drafted by Mentaj AI — refreshed today at 07:30 AM

UPCOMING SUBMISSIONS

Next 30 days — Ofsted, KHDA

Q2 Incident & safeguarding return

From: Head-02-04 — covers 1 Mar — 31 May 2026 — 4 incidents logged this quarter

Due 04 Jun 2026

Review draft

Generate & submit

Term 3 attendance summary

From: Head-071-12 — grades KST — Q3 — pre-filled from daily attendance data

Due 30 June

Review draft

Submit

Annual staffing & licensing roster

From: Head-071-04 — 30 staff — 3, 2 missing licence numbers (Sh. Reem - MR, Hassan) — click to fix

Due 16 July

Fix & preview

Generate

GENERATE A REPORT

Pre-formatted for the selected regulator

Enrolment by grade

Use headcount per grade with bilingual labels and gender breakdown.

Last gen 14 May

PDF

CSV

Attendance summary

Daily, weekly, or term — per student or aggregated — regulator required table.

Last gen 2 May

PDF

XLSX

Staff roster & licences

Teaching staff with licence numbers, expiry dates, qualifications, contract type.

Last gen 28 Apr

PDF

XLSX

Demographics breakdown

Nationality, language, special needs flags — in both standard categories.

Last gen 14 Apr

PDF

CSV

Safeguarding & incidents

Logged incidents with response actions — restricted to student priority.

Last gen 30 Apr

PDF

Curriculum alignment

Subjects taught vs MOC curriculum framework — gaps highlighted.

Last gen 28 Apr

PDF

RECENT SUBMISSIONS

[View full history](#) →

REPORT	PERIOD	SUBMITTED	BY	STATUS
Term 2 attendance summary	1 Jan — 28 Mar 2026	30 Apr 2026	Dr. Patel	ACCEPTED
Q1 Incident & safeguarding return	1 Jan — 31 Mar 2026	10 Apr 2026	Ms. Salwa	ACCEPTED
Staffing change notice — new hire	15 Apr 2026	16 Apr 2026	HR	ACCEPTED
Term 1 attendance summary	1 Sep — 18 Dec 2025	10 Jan 2026	Dr. Patel	ACCEPTED
Annual enrolment declaration	2025/26 academic year	5 Oct 2025	Dr. Patel	ACCEPTED

COMPLIANCE SNAPSHOT

Live — refreshed daily

CURRICULUM ALIGNMENT

✓ **Aligned**

All grades — last audit 28 Mar

TEACHER LICENSING

➡ **2 expiring soon**

Both renewals due before 30 Jun

SAFEGUARDING TRAINING

➡ **100% complete**

All 50 staff certified this year

HEALTH & SAFETY AUDIT

➡ **Passed**

Last inspection 12 Feb — next due Sep 2026

[← Back to all messages](#) · Starting message library v1.0

Fee/invoice reporting note (X20 → Parent P5): the fee-delinquency / AR console is **moved to the Parent archetype (P5, invoices)** for Phase 1; an admin-side AR view is Phase 2. Shown for context:

Fee delinquency & accounts receivable

All outstanding balances across the school. Payments happen in the school's portal — we track, chase, and report.

M

3 accounts crossed 60 days this week and need a phone call — not another email. Total AR is down 8% vs last month thanks to the early-installment-discount cycle. One family (Al-Lawati) sits on both this list and the retention dashboard — coordinate with admissions before the next nudge. Drafted by Manhaj AI - refreshed today - uses 90-day chase-cycle history

TOTAL OUTSTANDING AR

AED 145.2k

5.3% of monthly billing - 51 accounts

VS LAST MONTH

-8%

AED 12.6k collected - 9 accounts cleared

ACCOUNTS OVERDUE

21 / 51

past their 15-day soft window

OLDEST DEBT

78 days

1 G12 leaver - final-term balance - escalated

Aging buckets

Where the AED 145.2k sits today.

0 — 15 days

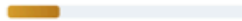
Current / soft

**AED 78.2k**

30 accounts

16 — 30 days

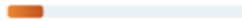
Chase

**AED 32.1k**

12 accounts

31 — 60 days

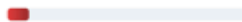
Escalate

**AED 22.1k**

6 accounts

60+ days

Board level

**AED 12.8k**

3 accounts

AR by grade

Top 8 grades by outstanding - click to filter.



NEEDS YOU THIS WEEK

5 actions, ordered by priority



3 accounts crossed 60 days

Auto-escalation rule triggered. These need a phone call — emails won't move them now.

[Open list](#)

Maya Al-Lawati — coordinate with admissions

32 days overdue - also on retention high-risk - soft check-in, not chase. Admissions has a plan.

[Retention plan](#)[Draft soft check-in](#)

12 routine 15-day reminders

Soft, first-touch reminders. AI-drafted in the school's voice. Preview before sending.

[Preview all](#)[Send batch](#)

1 returned cheque

Al-Rashdi family - cheque bounced 2 days ago - template ready for redeposit request.

[Open record](#)

G12 leaver — final balance

Faisal Al-Kindi - AED 4.8k outstanding - 78 days - graduating 25 Jun.

[Escalate to principal](#)

Monthly AR report

For the board pack - cash position, aging, vs last 3 months.

[Generate PDF](#)

ALL OUTSTANDING BALANCES

51 accounts - sorted by aging x amount

All

60+ days

31-60

16-30

0-15

[On retention list](#)

FAMILY	CHILD(REN)	AMOUNT	AGE	LAST CONTACT	REMINDER STAGE	ACTIONS
Faisal Al-Kindi family 1 child - graduating	Faisal - G12	AED 4,800	78d	Call attempt 4 days ago - no answer	ESCALATED	Call log Pay portal > Escalate
Al-Lawati family 1 child - 4yr ISO	Maya - G5	AED 8,750	32d	Reminder #3 3 days ago - unread	R3 + HELD retention	Soft check-in Retention plan Pay portal >
Al-Mahri family 1 child - 2yr ISO	Khaili - G7	AED 8,500	18d	Reminder #2 5 days ago - read	R2 + retention	Draft R3 Retention plan Pay portal >
Al-Rashdi family 2 children - 6yr ISO	Fatima - G4 Nasser - G6	AED 17,200	22d	Cheque bounced 2 days ago	CHEQUE RETURNED	Open record Redeposit request Pay portal >
Al-Said family 2 children - 5yr ISO	Hassan - G3 Sara - G7	AED 14,400	17d	Reminder #2 1 day ago - opened	R2	Draft R3 Pay portal >
Al-Bulushi family 1 child - 6yr ISO	Aisha - G10	AED 11,200	19d	Reminder #2 2 days ago - read	R2 + retention	Coord with principal Pay portal >
Al-Hinal family 1 child - new family	Layla - KG1	AED 6,500	16d	Reminder #1 9 days ago - read	R2	Draft R2 Pay portal >
Al-Mughairi family 2 children - 3yr ISO	Salim - KG2 Hanan - G3	AED 12,400	11d	no contact attempt yet	SOFT (R1)	Draft R1 Pay portal >
Al-Maamari family 1 child - 4yr ISO	Zayn - G6	AED 7,800	8d	no contact attempt yet	SOFT (R1)	Draft R1 Pay portal >

+ 42 more accounts - see all

Action buttons marked > open the school's payment portal in a new tab. Reminders are drafted by AI but always reviewed before sending. Stage rules: R1 (soft) → R2 (chase) → R3 (formal) → Escalated (principal/board).

Fee Delinquency & AR

Screen mockup of the Fee Delinquency & AR system, showing the main dashboard and key features.

Visual summary Backend & data The mockup

Open mockup full-screen ↗

1 - VISUAL SUMMARY

What this screen is

Bursar's view of all outstanding balances, aging buckets, by-grade AR distribution, and chase actions — with payment links that redirect to the school's payment portal.

WHO IT'S FOR

Bursar, business manager, or principal at smaller schools.

WORKFLOW IT SUPPORTS

Replaces spreadsheet chase lists. Surfaces patterns (aging by grade, repeat offenders). Coordinates with retention dashboard for families flagged on both lists.

KEY FEATURES

- Total AR with vs last-month trend
- Aging buckets visualization (0–15 / 16–30 / 31–60 / 60+ days)
- AR by grade — pattern detection (K12 seasonal, G7 a specific family)
- Action queue: 60+ day calls / soft check-ins for at-risk families / routine reminders / returned cheques
- Full AR table with reminder stages, retention cross-flag, deep-links to school's payment portal
- AI-drafted reminder generation per stage (K1 soft → K2 → K3 → escalated)
- Monthly AR report generation for board pack

2 - BACKEND & DATA

What it takes to build this for real

Data inputs, backend services, AI role, and privacy considerations.

INPUT DATA REQUIRED

- Invoices and payment status from school's billing system
- Communication history per family (which reminders sent, opened, ignored)
- Cross-reference with retention dashboard (which families are also flagged academically)
- Aging-metadata (days since invoice issued, days since due)

BACKEND BUILDS REQUIRED

- One-way sync from school's billing system (read-only)
- Reminder template library with stage rules (K1 → soft, K2 → chase, K3 → formal, Escalated → principal)
- AI message generation in school's voice per stage
- Per-family communication log with delivery → open status
- Cross-link with retention dashboard (shared family table)
- PDF generation for monthly AR report
- Deep-link generation to school's payment portal with parent ID context

AI ROLE

Medium: AI drafts reminder messages in school's voice with appropriate tone per stage. Pattern detection on AR by grade.

PRIVACY & ACCESS

Finance team → principal access. Cross-links to retention dashboard respect role gating (Bursar sees the financials; principal sees both failures).

3 - THE MOCKUP

Live preview

Embedded below — or open full-screen ↗ to interact with it directly.

Mockup file: admin-fee-delinquency.html Full-screen ↗

Marhalj
Marhalj Agency

Dashboard Students Admissions Finance Reporting

Bursar Mrs. Ameer

Fee delinquency & accounts receivable

All outstanding balances across the school. Payments happen in the school's portal — we track, chase, and report.

3 accounts crossed 60-days this week and need a phone call — not another email. Total AR is down 8% vs last month thanks to the early installment-discount cycle. One family (Al-Lawati) sits on both this list and the retention dashboard — coordinate with admissions before the next nudges.
Drafted by Marhalj AI - informed today · uses 60-day chase cycle history

TOTAL OUTSTANDING AR
AED 145.2k
5.8% of monthly billing · 8% monthly

VS LAST MONTH
-8%
AED 12.8k collected · 9 accounts cleared

ACCOUNTS OVERDUE
21 / 81
past their 15-day soft window

OLDEST DEBT
78 days
1 (10% owed - final term balance overdue)

Aging buckets
where the AED 145.2k sits today

Bucket	Accounts	Amount
0 – 15 days Current / soft	30 accounts	AED 78.2k
16 – 30 days Chase	10 accounts	AED 32.1k
31 – 60 days Escalate	8 accounts	AED 23.1k
60+ days Board level	3 accounts	AED 12.8k

AR by grade
Top 8 grades by outstanding · click to filter

Grade	Amount
K12	AED 24.6k
G7	AED 22.1k
G6	AED 18.0k
K01	AED 16.5k
K2	AED 16.1k
G12	AED 13.5k
G8	AED 12.0k
G10	AED 9.0k

5 actions, ordered by priority

3 accounts crossed 60-days
Auto-escalation rule triggered. Please read a phone call — email won't move them now.
[Open list](#)

Days Al-Lawati — coordinate with admissions
30 days overdue · also on retention high-risk · soft check-in, not chase. Admissions has a plan.
[Retention plan](#) [Draft soft check-in](#)

12 routine 15-day reminders
Soft, 15-day reminders · all drafted in the school's voice. Preview before sending.
[Preview all](#) [Send batch](#)

1 returned cheque
Al-Badair family · cheque bounced 2 days ago · template ready for redisplay request.
[Open request](#)

G12 learner — final balance
Faisal Al-Kindi · AED 4.8k outstanding · 18 days · graduating 28 Jun.
[Escalate to principal](#)

Monthly AR report
For the board pack · cash position, aging, vs last 3 months.
[Download PDF](#)

ALL OUTSTANDING BALANCES

87 accounts · sorted by aging → amount

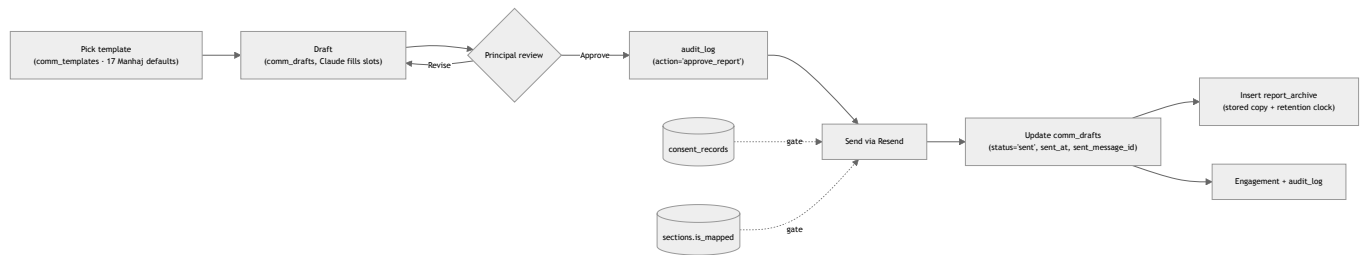
Search by family, class, or invoice ID

FAMILY	CHILDREN	AMOUNT	AGE	LAST CONTACT	REMARKS / STAGE	ACTIONS
Faisal Al-Kindi Family 1 child graduating	Faisal · G12	AED 4,800	78d	Call attempt 4 days ago · no answer	ESCALATE	Call log Flag portal Escalate



Who uses it & permissions. Admin/Principal (approve + send). Advisor (draft only). Sends are **consent-gated** and **section-mapping-gated** (A3).

Workflow (parent-comms pipeline).



Backend & data linkage.

- Templates come from **comm_templates** (17 Manhaj defaults seeded). Drafting writes **comm_drafts** (Claude fills the slots). The **comm_draft_status** enum has **no approved value** — it only goes **draft** → **sent**. Principal approval is recorded in **audit_log** (**action='approve_report'**), **not** as a draft-status change; sending then flips **comm_drafts.status** to **sent**, calls **Resend**, and records **sent_at** + **sent_message_id**.
- **Consent gate:** before send, check **consent_records** (PDPL/DPL compliance). **Section-mapping gate:** unmapped sections can't send (A3). Every send writes **audit_log**.
- **Report storage + tracking + retention (X16 — Phase 1).** Add **NEW report_archive** (**id, school_id, report_kind, scope, storage_path, generated_at, sent_at, delete_after, deleted_at**) to **store past reports, track** what was produced/sent, and **delete after 2–3 years** via a retention job that respects **delete_after**. **The link to the parent piece is key:** a parent-facing report (digest/invoice/comms) that is sent is archived here with the same **student_id / parent_id** so "what did this family receive, and when" is answerable — this is the bridge to **Parent P1 (weekly digest)** and **Parent P5 (invoices)**.
- **Assessment/exam oversight (X13 — Phase 1, read-only).** A **NEW read-only admin view** **v_assessment_oversight** over existing **assessments + assessment_results** so the principal can see quiz/exam scores organised by **student name, id, class, grade, completed-exam** — no new write path; teachers still own the scores. This relates directly to **assessments / assessment_results**.
- **Eval storage + the teacher role (X17/X18).** Per-child satisfaction/eval rollups rely on a **deferred parent-feedback surface** (X17) that does not yet exist — **Parent P1 (weekly digest) only pushes the digest out; it captures no parent evaluations**, so there is no Phase-1 source for these rollups (correctly Phase 2). The eval-creation flow (X18) is where a **sign-up gives a role to the teacher** (the teacher who creates/owns a rubric/eval) — cross-link the **Teacher rubric/eval** section; admin only needs the read/aggregate, stored in table form (the eval rows the teacher writes, once a feedback surface exists to generate them).
- **Regulatory reports (BLOCKED — research first).** After the MoE research lands, add **NEW regulatory_report_catalog** (**report_type, regulator, required_fields, deadline_cadence, template_ref**) and the generators that fill each template from the spine.

Data tables.

comm_drafts (existing) — a parent-comms send. *Names in **student** / **parent** are display stand-ins for the uuid FK.*

template	student	parent	status	sent_at	sent_message_id
absent_today	Khalil Al-Mahri (G5)	Mr Al-Mahri	sent	2026-05-31T08:10:00Z	resend_a91f...
well_done_milestone	Layla Al-Habsi (G10)	Mr Al-Habsi	draft	—	—

NEW report_archive (X16) — stored reports + retention clock:

report_kind	scope	storage_path	generated_at	sent_at	delete_after
parent_weekly_digest	Layla Al-Habsi · wk 2026-05-25	storage://reports/.../digest.pdf	2026-05-29T18:00:00Z	2026-05-29T18:05:00Z	2029-05-29
moe_enrolment_return	term 2026-T3	storage://reports/.../moe.pdf	2026-06-15T10:00:00Z	2026-06-15T11:30:00Z	2029-06-15

NEW `v_assessment_oversight` (X13, read-only view over existing tables). Names in the `student` column are display stand-ins for the `uuid` FK.

student	external_ref	class	grade	assessment (completed exam)	score / max
Maya Al-Lawati	STU-2461	G5B	5	Term 3 Maths exam	88 / 100
Khalil Al-Mahri	STU-2452	G5B	5	Term 3 Maths exam	61 / 100

NEW `regulatory_report_catalog` (post-MoE-research) — illustrative:

report_type	regulator	deadline_cadence	template_ref
enrolment_return	Oman MoE	termly	(TBD — sourced from MoE)
attendance_summary	Oman MoE	monthly	(TBD — sourced from MoE)

Deferred to P2/P3.

- Regulatory report generators** — *blocked, not deferred by choice*: cannot build the generators until the MoE templates/content/deadlines are sourced (research task in §Z). *Clean-linkage*: the `regulatory_report_catalog` + `report_archive` tables are scaffolded in P1 so the generators slot in without migration.
- Parent-satisfaction analytics** (`admin-parent-satisfaction`) → Phase 2 — depends on a **deferred parent-feedback surface (X17)** existing first to have data to aggregate. Parent P1 (weekly digest) is a one-way send and captures **no evaluations**, so there is no Phase-

1 data source — correctly Phase 2.



Open questions.

- **MoE region first** (Oman vs UAE) and whether any templates are already in hand (sales to source — the blocker).
- Retention window for `report_archive.delete_after` — confirm 2 vs 3 years per regulator/PDPL (X16).
- **Report-view validation (X23)**: which report views does the school actually want, and which need custom/better versions — research/validation task owned by Sales (with CTO), reviewed internally too.

Z. Build & data handover

Task allocation

Build item	Owner role	Why
Repo split into 4 apps + shared core / auth / ui / schema packages	Engineer	Foundation; one schema, zero drift.
Auth (magic-link onboarding, password login, role guard) + Manage Admins (invitations , school_admins)	Engineer	Gates every screen; RLS + guard wiring.
Dashboard cron AI briefing + ai_briefings + count cards	Engineer	Scheduled Claude job + cached read.
Input Data CRUD (Students/Teachers/Parents) + Section Mapping save	Engineer	Typed server actions + validation; the core write surface.
Faculty HR (roster/contracts/staffing category) + sick-leave approval state machine (staff_absences)	Engineer	Approval is the trigger for Teacher T6.
Students: at-risk routing (risk_flags + RLS), re-enrolment funnel, leavers, applicant pipeline + convert(applicants)	Engineer	Biggest SIS write-path gap.
Attendance aggregation views + sick-leave intake wiring	Engineer	Read-only rollups over attendance_marks ; shared staff_absences .
Schedule + CP-SAT solver service + NL-constraint parse + substitute finder	Engineer	Heaviest item; separate FastAPI service.
Reports comms pipeline + report_archive (X16) + v_assessment_oversight (X13)	Engineer	Draft→approve→send + storage/retention + oversight view.
CP-SAT service infra + AI cost sizing (dashboard, schedule, comms drafts)	Hybrid CTO	Architecture + per-tenant cost cap decisions.
At-risk owner decision (advisor / homeroom / both + precedence)	Hybrid CTO	Shapes RLS + the teacher app touch-point.
MoE regulatory research (templates, mandated content, deadlines; region first)	Sales (non-technical)	The Reports blocker; needs direct regulator/school access.
Report-view validation with the school (X23)	Sales (non-technical) + CTO	Confirm which report views are wanted / need custom versions.
Source staffing-category list, leaver reasons, substitute-cover policy, payment-portal details	Sales (non-technical)	School holds these; needed before A3/A4/A7/Parent-P5 are trustworthy.

Data to source from the school

Table / dataset	Fields needed	Source (who/where at school)	Why it's needed / what it feeds
MoE regulatory templates	mandated report types, required content/format, deadlines , region (Oman/UAE)	Principal / compliance officer / Ministry	Unblocks A8 regulatory reports ; fills <code>regulatory_report_catalog</code> .
Staffing categories	the school's own workload categories	HR / principal	A3/A4 <code>staffing_category</code> ; workload grouping.
Leaver reasons	the school's withdrawal-reason vocabulary	Registrar / admissions	A5 <code>students.withdrawn_reason</code> ; feeds MoE leaver reporting.
Substitute-cover policy	how subs are ranked/assigned after sick-leave approval	Operations head / deputy	A7 substitute finder weights + the Teacher T6 trigger.
Applicant pipeline stages	the school's real admissions stages + source channels	Admissions officer	A5 <code>applicants</code> stage list + conversion.
Parent contacts + consent	email, preferred language, consent kind/granted	Registrar / parent forms	A8 comms routing + the consent gate (PDPL).
Report retention rule	how long past reports must be kept (2 vs 3 yrs)	Compliance / regulator	A8 <code>report_archive.delete_after</code> (X16).
Payment-portal details	the school's payment-link/redirect target	Bursar / business manager	Parent P5 invoices (X20 moved); admin AR view (P2).
Dashboard cadence	daily vs 2–3x/week	Principal	A2 cron schedule + AI cost.

End of Admin (Principal) Phase-1 handover. NEW data objects introduced/confirmed here: `invitations` , `school_admins` , `ai_briefings` , `staff_absences` , `staffing-category` field on `teachers` , `applicants` , `risk_flags` , `students.advisor_id` , `students.withdrawn_reason` , `report_archive` (X16), `v_assessment_oversight` (X13), `regulatory_report_catalog` (post-MoE-research). Folded-in founder notes: X13 (exam oversight), X16 (report storage/tracking/retention), X17 (satisfaction depends on Parent P1 digest evals), X18 (eval storage + teacher-role cross-link), X19 (data + backend confirmed), X20→Parent P5, X21→Parent P2/P3 (deferred), X23 (report-view validation → Sales). Deferred: bulk import (P2), Performance Composite (P2–3), full safeguarding case mgmt (P2), reception pickup (P2), parent-satisfaction analytics (P2), admin AR console (P2); regulatory generators BLOCKED on MoE research.